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Improvements to the post-processing of weather forecasts using machine learning and feature selection

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ABSTRACT

This study aims to develop and improve machine learning-based post-processing models for precipitation, temperature, and wind speed predictions using the Mesoscale Model (MSM) dataset provided by the Japan Meteorological Agency (JMA) for 18 locations across Japan, including plains, mountainous regions, and islands. By incorporating meteorological variables from grid points surrounding the target locations as input features and applying feature selection based on correlation analysis, we found that, in our experimental setting, the LightGBM-based models achieved lower RMSE than the specific neural-network baselines tested in this study, including a reproduced CNN baseline, and also generally achieved lower RMSE than both the raw MSM forecasts and the JMA post-processing product, MSM Guidance (MSMG), across many locations and forecast lead times. Because precipitation has a highly skewed distribution with many zero cases, we additionally examined Tweedie-based loss functions and event-weighted training strategies for precipitation forecasting. These improved event-oriented performance relative to the original LightGBM model, especially at higher rainfall thresholds, although the gains were site dependent and overall performance remained slightly below MSMG.

1. Introduction

1.1. Research background and objectives

Numerical weather prediction (NWP) is a primary method of weather forecasting that represents future atmospheric states using mathematical models based on observation data. Over the years, it has evolved through advancements in the representation of physical processes, model initialization, including assimilation, and the introduction of ensemble modeling (Bauer et al., 2015). However, NWP is not without its limitations, such as forecast errors and biases. While forecast variance might be unavoidable, several post-processing techniques using machine learning models have been developed to mitigate these biases (Liu et al., 2023; Rojas-Campos et al., 2023; Yoshikane and Yoshimura, 2022; Zhang and Ye, 2021; Kudo, 2022; Peng et al., 2020; Tang et al., 2021; Salazar et al., 2022; Xu et al., 2020).

The Japan Meteorological Agency (JMA) has also introduced a post-processing method called the MSM Guidance (MSMG) to correct the errors of the Mesoscale Model (MSM), which has a finer grid spacing (5 km) and targets areas around Japan compared to global models (JMA, 2024). In MSMG, JMA primarily employs statistical methods, such as the Kalman filter, frequency bias correction, and neural networks, to reduce the systematic biases in MSM.

This study aims to develop machine learning-based post-processing models for precipitation, temperature, and wind speed predictions – representative meteorological variables – using MSM data for 18 locations across Japan, including plains, mountainous regions, and islands. Meteorological variables from grid points surrounding the prediction locations were used as input features, and feature selection based on correlation analysis was applied. In our experimental setting, the LightGBM-based models achieved lower RMSE than the tested neural-network baselines, including the reproduced CNN baseline, and also generally achieved lower RMSE than both the raw MSM forecasts and the JMA post-processing product, MSM Guidance (MSMG). Among the LightGBM-based models, those using surrounding-grid information with correlation-based feature selection showed the lowest RMSE across many locations and forecast lead times.

For precipitation, we further examined Tweedie-based loss functions and event-weighted training strategies, which improved event-oriented metrics for some sites and rainfall thresholds even when the gains in RMSE were limited.

1.2. Related work

Several studies have developed post-processing models for precipitation using methods such as convolutional neural networks (CNNs),

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利用 机器学习 和 特征选择 对 天气预报 的 后处理 进行 改进 与 优化

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ABSTRACT

本研究旨在开发并改进基于机器学习的后处理模型，用于降水、温度和风速预测，使用日本气象厅提供的日本18个地点（包括平原、山地地区和岛屿）的中尺度模式数据集。通过将目标地点周围格点的气象变量作为输入特征，并基于相关分析进行特征选择，我们发现，在我们的实验设置中，基于LightGBM的模型比本研究中测试的特定神经网络基线（包括复现的CNN基线）实现了更低的均方根误差，并且在许多地点和预报时效上，通常也比原始MSM预报和JMA后处理产品MSM指导产品取得了更低的均方根误差。由于降水分布高度偏斜且包含大量零值，我们还额外研究了基于Tweedie的损失函数和事件加权训练策略用于降水预报。这些方法相对于原始LightGBM模型提高了事件导向的性能，尤其是在较高降雨阈值下，尽管增益因地点而异，且整体性能仍略低于MSM指导产品。

1. 引言

1.1. 研究 背景 与 目标

数值天气预报（NWP）是一种基于观测数据，利用数学模型表示未来大气状态的主要天气预报方法。多年来，它通过在物理过程描述、模式初始化（包括同化）以及引入集合建模方面的进步而不断发展（Bauer等，2015）。然而，数值天气预报并非没有局限性，例如预报误差和偏差。尽管预报方差可能无法避免，但已经开发了几种使用机器学习模型的后处理技术来减轻这些偏差（Liu等，2023；Rojas-Campos等，2023；Yoshikane和Yoshimura，2022；Zhang和Ye，2021；Kudo，2022；Peng等，2020；Tang等，2021；Salazar等，2022；Xu等，2020）。

日本气象厅还引入了一种名为MSM指导产品（MSMG）的后处理方法，用以修正中尺度模式（MSM）的误差。与全球模式相比，MSM具有更精细的网格间距（5公里），并覆盖日本周边地区（JMA，2024）。在MSMG中，日本气象厅主要采用统计方法，如卡尔曼滤波、频率偏差修正和神经网络，以减少MSM中的系统性偏差。

本研究旨在利用机器学习开发后处理模型，针对降水、温度和风速——这三种代表性的气象变量——使用MSM数据对日本全境18个地点（包括平原、山地地区和岛屿）进行预测。预测地点周围格点的气象变量被用作输入特征，并应用了基于相关分析的特征选择。在我们的实验设置中，基于LightGBM的模型比测试的神经网络基线（包括复现的CNN基线）实现了更低的均方根误差，并且通常也比原始MSM预报和日本气象厅后处理产品MSM指导产品（MSMG）实现了更低的均方根误差。在基于LightGBM的模型中，使用周围网格信息并采用基于相关特征选择的模型在多个地点和预报时效中显示出最低的均方根误差。

对于 降水， 我们 进一步 检验了 基于Tweedie的 损失 函数 和 事件加权 训练 策略， 这些策略 在 RMSE 提升 有限 的情况下， 仍 改善了 某些站点 和 降雨阈值 的 事件导向 指标。

1.2. 相关工作

多项 研究 已 开发 出 后处理 模型 用于 降水 预测 ， 采用 诸如卷积 神经 网络(CNN)等方法,

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neural networks, and Support Vector Machines (Zhang and Ye, 2021; Liu et al., 2023; Rojas-Campos et al., 2023; Yoshikane and Yoshimura, 2022). Among them, Zhang and Ye (2021) conducted comparative experiments on machine learning models, input parameters, and training data periods for precipitation prediction, and concluded that LightGBM provides the most balanced performance among the tested models.

For temperature post-processing models, CNNs, neural networks, and LightGBM have been employed (Kudo, 2022; Peng et al., 2020; Tang et al., 2021). Kudo (2022) developed a CNN-based temperature post-processing model targeting the Kanto region of Japan and demonstrated its superior performance over the JMA's MSMG.

Wind speed post-processing models have also been explored using neural networks and LightGBM (Salazar et al., 2022; Tang et al., 2021; Xu et al., 2020). Xu et al. (2020) utilized LightGBM for wind speed prediction, analyzing feature importance derived from the model. Their study revealed that using all features, including various weather elements, as input outperformed models where features were limited primarily to wind speed.

In addition to using NWP outputs, some studies have leveraged meteorological forecast data from existing weather forecasting services (Iwase and Takenawa, 2024; Tsipis et al., 2023). Iwase and Takenawa (2024) employed machine learning models such as LightGBM, XGBoost, and neural networks to predict temperature and precipitation in mountainous regions. By incorporating surrounding weather forecast data as input variables, the results were superior to existing weather forecast services.

In recent machine learning research, deep learning methods have been developed significantly in fields such as natural language processing and image recognition. In contrast, for tabular data, tree-based models currently outperform deep learning models in terms of accuracy and require less hyperparameter tuning costs (Shwartz-Ziv and Armon, 2022; Grinsztajn et al., 2022). The utility of tree-based models has also been demonstrated in post-processing NWP outputs, which often involve tabular data (Zhang and Ye, 2021; Xu et al., 2020). Recently, Hieta and Partio (2025) reported that a tree-based gradient-boosting post-processing scheme (XGBoost) reduces RMSE for short-range near-surface forecasts.

The remainder of this paper is organized as follows. Section 2 defines the methods and data used in this study. Section 3 presents the results and discussion. Finally, Section 4 concludes the study.

2. Methods

2.1. Data

In this study, we use MSM data from the JMA as the input for the model and observation data as the target data for training. The MSM data were collected and distributed by Research Institute for Sustainable Humanosphere, Kyoto University (RISH, 2024). The observation data from JMA include cumulative precipitation over the past three hours, hourly temperature, and wind speed. We selected the 18 observation sites from all over Japan (Fig. 1, Table 1). Additionally, in order to compare the forecast results, we collected MSMG data, a post-processed forecast provided by JMA and distributed by the Japan Meteorological Business Support Center (JMBSC).

MSM consists of three-dimensional gridded predictions of future atmospheric conditions, including temperature, wind, moisture, and solar radiation, with fine grid spacing of approximately 5 km. The forecasts cover Japan and surrounding sea areas (JMA, 2024). Surface-level data are available at hourly intervals, while pressure-level data are available at three-hour intervals. The forecasts are issued every three hours (00, 03, 06, 09, 12, 15, 18, 21 UTC) and extend up to 39 h ahead. The surface grid comprises 505×481 points (grid spacing: 0.05° latitude × 0.0625° longitude), while the pressure-level grid consists of 253×241 points (grid spacing: 0.1° latitude × 0.125° longitude),

Table 1
Latitude, longitude, altitude, and station numbers of the 18 sites shown in Fig. 1.

	Latitude	Longitude	Altitude	Station number
Ashimine	26.193	127.638	3	0
Uchinoura	31.277	131.055	8	1
Minamiaso	32.832	131.013	394	2
Uwanokogen	35.432	134.583	540	3
Imabari	34.053	132.975	27	4
Sakai	34.555	135.485	20	5
Kuwana	35.050	136.693	3	6
Sekigahara	35.363	136.467	130	7
Itoigawa	37.043	137.875	8	8
Nobeyama	35.948	138.472	1350	9
Niijima	34.368	139.268	29	10
Tsujido	35.320	139.450	5	11
Saitama	35.875	139.587	8	12
Kusatsu	36.617	138.592	1223	13
Kamaishi	39.270	141.878	5	14
Oma	41.527	140.912	14	15
Ishikari	43.193	141.370	5	16
Kamikawa	43.847	142.753	324	17

Table 2
Meteorological elements used as predictors, including names and units. A ✓indicates that the element is used at the surface and/or at the representative pressure levels (850, 500, and 200 hPa). (Relative humidity is used at 850/500 hPa only.)

Element name	Surface (hourly)	Pressure level (every 3 h)	Unit
Temperature	✓	✓	°C
Relative humidity	✓	850/500 hPa only	%
Hourly precipitation	✓		mm
Wind speed	✓	✓	m s^{-1}
U wind component	✓	✓	m s^{-1}
V wind component	✓	✓	m s^{-1}
Vertical velocity		✓	hPa s^{-1}
Surface pressure	✓		hPa
Sea level pressure	✓		hPa
Geopotential height		✓	m
Solar radiation	✓		W m^{-2}
Total cloud cover	✓		%
Low cloud cover	✓		%
Mid-level cloud cover	✓		%
High cloud cover	✓		%

covering the area from the northwest corner (47.6° N, 120° E) to the southeast corner (22.4° N, 150° E).

MSM outputs a variety of meteorological elements on the surface and on multiple standard pressure levels. In this study, we use the elements listed in Table 2 as predictors. For pressure-level variables, although MSM provides outputs at more pressure levels, we use only three representative levels (850, 500, and 200 hPa), following prior work (Liu et al., 2023), to capture information from the lower, middle, and upper atmosphere while reducing dimensionality. The CNN baseline uses a different input configuration following prior work (including additional near-surface pressure levels); see Section 2.5.

For each station, we train an independent regression model. Except for the CNN-based model, the inputs to LightGBM and the neural network are one-dimensional feature vectors constructed by concatenating (i) surface variables within a temporal window $(t, t-1, t-2)$, (ii) pressure-level variables at the target time t only (850, 500, and 200 hPa), and (iii) auxiliary scalar features (lead time, month, and hour of day). All gridded predictors are flattened before concatenation. The temporal window length was chosen to match the 3-hourly forecast cycle of MSM/MSMG, so that a consistent set of predictors is available across lead times and forecast cycles.

The input variables consist of 13 surface variables and 7 pressure-level variables as summarized in Table 2. Among the pressure-level variables, relative humidity is available only at two levels (850 and

神经网络和支持向量机(Zhang and Ye, 2021; Liu等人，2023; Rojas-Campos 等人，2023; Yoshikane和Yoshimura，2022)。其中，Zhang and Ye (2021) 对降水预测的机器学习模型、输入参数和训练数据周期进行了对比实验，并得出结论：LightGBM在测试的模型中提供了最均衡的性能。

对于温度后处理模型，已采用CNN、神经网络和LightGBM（Kudo，2022；Peng等，2020；Tang等，2021）。Kudo（2022）开发了一个基于CNN的温度后处理模型，针对日本关东地区，并证明其性能优于日本气象厅的MSMG。

风速后处理模型也通过神经网络和LightGBM进行了探索（Salazar等，2022；Tang等，2021；Xu等，2020）。Xu等（2020）利用LightGBM进行风速预测，分析了从模型中得到的特征重要性。他们的研究表明，使用包括各种气象要素在内的全部数据作为输入，其性能优于特征主要限于风速的模型。

除了使用NWP输出，一些研究还利用了来自现有天气预报服务的气象预报数据（Iwase and Takenawa，2024；Tsipis等，2023）。Iwase and Takenawa（2024）采用机器学习模型，如LightGBM、XGBoost和神经网络，预测山区的温度和降水。通过将周边天气预报数据作为输入变量，其结果优于现有的天气预报服务。

在最近的机器学习研究中，深度学习方法在自然语言处理和图像识别等领域取得了显著发展。相比之下，对于表格数据，基于树的模型目前在准确性上优于深度学习模型，并且需要更少的超参数调整成本（Shwartz-Ziv and Armon, 2022；Grinsztajn 等人，2022）。树基模型在处理通常涉及表格数据的数值天气预报后处理中也被证明是有用的（Zhang and Ye, 2021；Xu 等人，2020）。最近，Hieta and Partio（2025）报告说，基于树的梯度提升后处理方案（XGBoost）降低了短程近地面预报的均方根误差。

本 文 的 其 余 部 分 结 构 如 下 。 第 2 节 定 义 了 本 研 究 所 用 的 方 法 和 数 据 。 第 3 节 展示结果与讨论。最后，第4节总结本研究。

2. 方法

2.1. 数据

在本研究中，我们使用日本气象厅的MSM数据作为模型输入，并将观测数据作为训练的目标数据。MSM数据由京都大学可持续人类圈研究所（RISH, 2024）收集和分发。日本气象厅的观测数据包括过去三小时的累积降水量、每小时温度以及风速。我们选择了日本全国18个观测站点（图1，表1）。此外，为了比较预报结果，我们还收集了日本气象厅提供并由日本气象业务支援中心（JMBSC）分发的后处理预报MSMG数据。

MSM 包含未来大气状况的三维网格预测，包括温度、风速、湿度和太阳辐射，网格间距精细约为 5 公里。预报覆盖日本及周边海域（JMA，2024）。地面数据每小时提供一次，而气压层数据每三小时提供一次。预报每三小时发布一次（00、03、06、09、12、15、18、21 UTC），最长可预测 39 小时。地面网格包含505×481 个点（网格间距：0.05° 纬度 × 0.0625° 经度），而气压层网格包含 253×241 个点（网格间距：0.1° 纬度 × 0.125° 经度）。

表 1 图1中显示18个站点的纬度、 经度、 海拔、 和 站点 编号 的 18 个 站点 如 图 1 所示。

	纬度	经度	海拔	站点 编号
Ashimine	26.193	127.638	3	0
Uchinoura	31.277	131.055	8	1
Minamiaso	32.832	131.013	394	2
Uwano kogen	35.432	134.583	540	3
今治	34.053	132.975	27	4
堺	34.555	135.485	20	5
桑名	35.050	136.693	3	6
Sekigahara	35.363	136.467	130	7
Itoigawa	37.043	137.875	8	8
Nob惠山	35.948	138.472	1350	9
Niijima	34.368	139.268	29	10
辻堂	35.320	139.450	5	11
Saitama	35.875	139.587	8	12
Kusatsu	36.617	138.592	1223	13
釜石市	39.270	141.878	5	14
Oma	41.527	140.912	14	15
Ishikari	43.193	141.370	5	16
Kamikawa	43.847	142.753	324	17

表2 用作预测因子的气象要素，包括名称和单位。✓表示该要素在地面和/或代表气压层（850、500和200百帕）上使用。（相对湿度仅在850/500百帕层使用。）

元素 名称	地面 (每小时)	Pr 气压 层 (每 3 小时)	Unit
温度	✓	✓	°C
相对湿度	✓	850/500 百帕 仅	%
逐时降水	✓		mm
风速 速度	✓	✓	m s^{-1}
U 风分量	✓	✓	m s^{-1}
V 风速 分量	✓	✓	m s^{-1}
垂直 速度		✓	hPa 秒$^{-1}$
地面 气压	✓		hPa
海平面 气压	✓		hPa
位势 ial height		✓	m
太阳 辐射	✓		W m^{-2}
总云 量	✓		%
低云 量	✓		%
中云 量	✓		%
高云 量	✓		%

覆盖 从 西北角 (47.6° N， 120° E) 到 东南角 (22.4° N， 150° E) 的区域。

MSM在地面和多个标准气压层上输出多种气象要素。在本研究中，我们使用表2列出的要素作为预测因子。对于气压层变量，尽管MSM在更多气压层上提供输出，但我们仅使用三个代表性层次（850百帕、500百帕和200百帕），遵循先前的工作（Liu等，2023），以捕获低层、中层和高层大气的信息，同时降低维度。CNN基线使用了不同的输入配置，遵循先前的工作（包括额外的近地面气压层）；详见第2.5节。

对于每个站点，我们训练一个独立的回归模型。除基于CNN的模型外，LightGBM和神经网络的输入是通过拼接以下内容构建的一维特征向量：（i）时间窗口$(t, t-1, t-2)$内的地表变量，（ii）目标时刻t 仅有的气压层变量（850百帕、500百帕和200百帕），以及（iii）辅助标量特征（提前时间、月和小时）。所有网格化预测因子在拼接前都被展平。时间窗口长度选择为与MSM/MSMG的3小时一次预报周期匹配，以便在超前时间和预报周期内获得一致的预测因子集。

输入 变量 包括 13个 地面 变量 和 7个 气压层 变量， 如 表2所示 。在这些 气压层 变量中， 相对 湿度 仅 在 两个 层次 （850 和

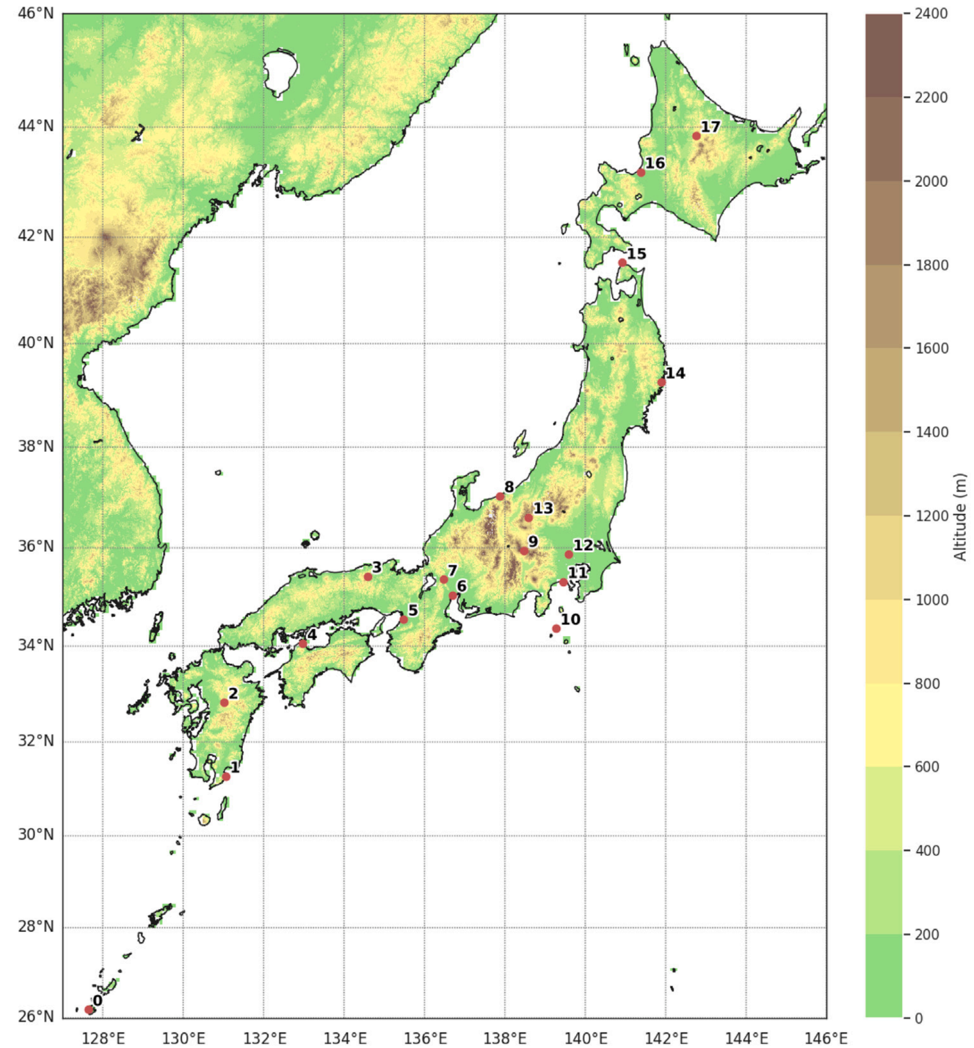

Fig. 1. Map showing the 18 selected locations.

500 hPa), whereas the other six variables are available at all three levels.

We prepared two input configurations. In the first, only the grid cell closest to the prediction point is used. In this configuration, the total number of input features is 62, computed as

$$13 \times 3 + (6 \times 3 + 2) + 3 = 62,$$

where 13×3 accounts for the 13 surface variables at three time steps, $(6 \times 3 + 2)$ counts pressure-level combinations (six variables at three levels plus relative humidity at two levels) at time t, and $+3$ accounts for lead time, month, and hour of day.

In the second, surrounding grid cells are included to assess the benefit of spatial context. Because the pressure-level fields have twice the horizontal grid spacing of the surface fields, we use an 11×11 grid for surface variables and a 7×7 grid for pressure-level variables so that the covered areas are comparable. Under this surrounding-grid configuration, the total number of input features is 5702, computed as

$$13 \times 3 \times 11 \times 11 + (6 \times 3 + 2) \times 7 \times 7 + 3 = 5702,$$

where 13 is the number of surface variables, 3 is the number of time steps for surface variables, 11×11 is the surface grid, $(6 \times 3 + 2)$ counts pressure-level combinations at time t, 7×7 is the pressure-level grid, and $+3$ accounts for lead time, month, and hour of day.

The output variables are the cumulative precipitation over the past three hours, as well as hourly temperature and wind speed. Observation data from 2019 to 2021 were used for training, 2022 for validation, and 2023 for testing, as shown in Table 3. Since the MSM provides complete fields without missing values, only observation samples containing missing values were excluded from the dataset. The validation data were used solely for model selection and hyperparameter tuning, and were not used to retrain the model prior to testing. Fig. 2 presents box plots for each location and meteorological variable.

For MSMG, the accumulated surface precipitation over the past three hours and hourly surface temperature and wind speed are predicted up to 39 h ahead, with forecast outputs available every three hours (00, 03, 06, 09, 12, 15, 18, and 21 UTC). Precipitation forecasts are provided on a grid with dimensions of 560 (longitude) × 480 (latitude), corresponding to a grid spacing of 0.05 ° in latitude and 0.0625 ° in longitude, covering the area from (47.975 °N, 120.03125 °E) in the northwest to (20.025 °N, 149.96875 °E) in the southeast. In contrast, forecasts for temperature and wind speed are provided at the locations of JMA observation stations.

The dataset is summarized as follows. The outputs consist of three variables (precipitation, temperature, and wind speed), while the inputs are identical across them. Missing values occur only in the observation data. The period covers the entire span of the training (2019–2021), validation (2022), and test (2023) sets. The total number of rows is determined by the combination of period, forecast cycles, lead times, and locations, while the number of columns differs according to the input configuration.

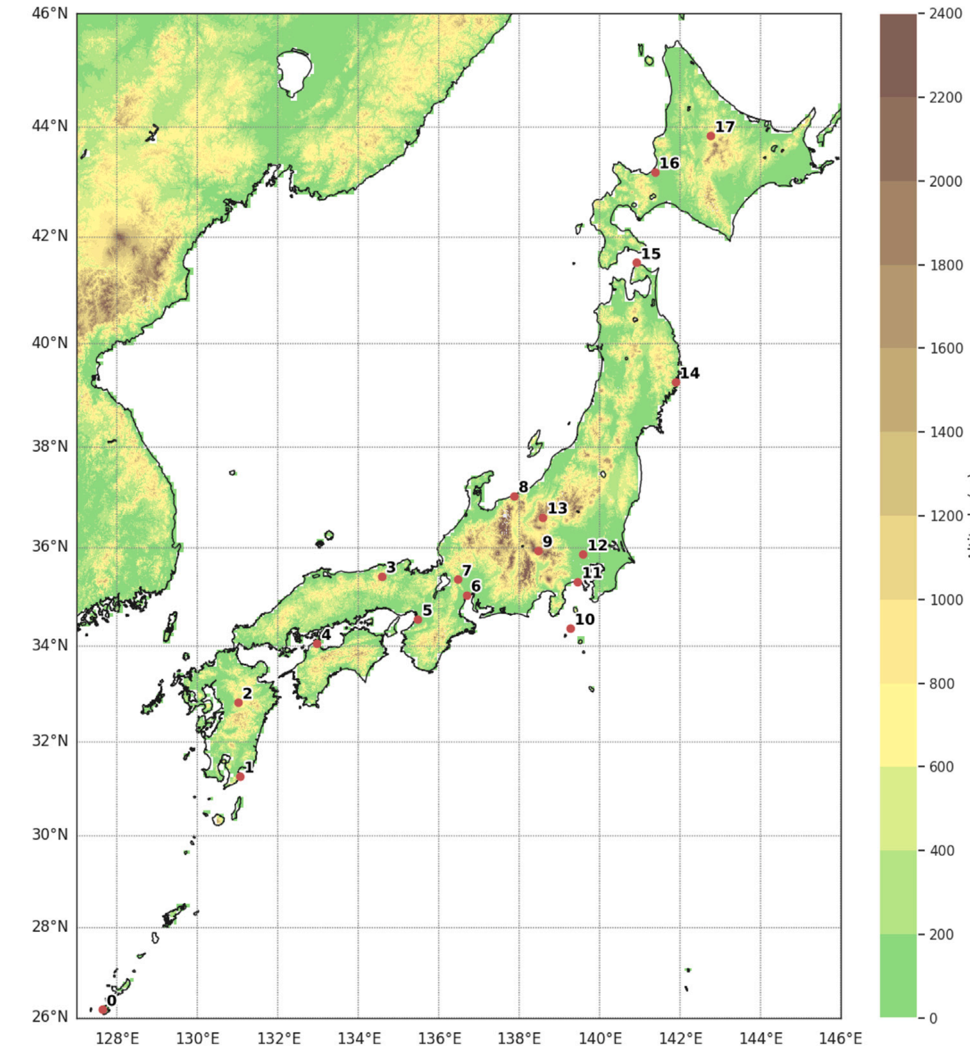

图 1. 展示 18 个 选定 地点 的地图。

500<hPa)， 而其他 六个 变量 在 全部数据 三个层次上均可用。

我们 准备了 两种 输入 配置。 在 第一种 配置中， 仅 使用 距离 预测 点 最近 的 网格 单元 。 在 这种 配置下， 输入 特征 的 总 数 为 62， 计算 方式 为

$$13 \times 3 + (6 \times 3 + 2) + 3 = 62,$$

其中13×3 代表三个时间步的13个地表变量，$(6 \times 3 + 2)$ 统计t时刻的气压层组合（六个变量在三个层次加上两个层次的相对湿度），+3 代表提前时间、月和小时。

在第二种配置中，包含周围的网格单元以评估空间上下文的好处。由于气压层场的水平网格间距是地表场的两倍，我们对地表变量使用 11×11 网格，对气压层变量使用7×7 网格，以便覆盖区域相当。在这种周围网格配置下，输入特征总数为5702，计算方式为

$$13 \times 3 \times 11 \times 11 + (6 \times 3 + 2) \times 7 \times 7 + 3 = 5702,$$

其中13是地表变量的数量，3是地表变量的时间步数，11×11 是地表网格，$(6 \times 3 + 2)$ 统计时间t上的气压层组合， 7×7 是气压层网格，+3 代表提前时间、月和小时。

输出 变量 是 过去 三 小时 的 累积 降水 量， 以及 每小时 的 温度 和 风速 。 观测数据来自2019年至2021年用于训练，2022年用于验证，2023年用于测试，如表3所示。由于MSM提供完整字段且无缺失值，仅从数据集中排除包含缺失值的观测样本。验证数据仅用于模型选择和超参数调整，在测试前未用于重新训练模型。图2展示了每个地点和气象变量的箱线图。

对于MSMG，过去三小时的累积地表降水量以及每小时的地表温度和风速，可提前39小时进行预测，预报输出每三小时一次（00、03、06、09、12、15、18和21 UTC）。降水预报在维度为560（经度）× × 480（纬度）的网格上提供，对应的网格间距为纬度0.05 °、经度0.0625 °，覆盖范围为西北方向（47.975 °N，120.03125 °E）至东南方向（20.025 °N，149.96875 °E）。相比之下，温度和风速的预报则在JMA观测站的地点提供。

数据集总结如下。输出包含三个变量（降水、温度和风速），而输入在各变量中相同。缺失值仅出现在观测数据中。时间范围涵盖整个训练集（2019–2021）、验证集（2022年）和测试集（2023年）。总行数由时段、预报周期、超前时间和地点的组合决定，而列数则根据输入配置有所不同。

Table 3
The number of samples in the training, validation, and test datasets, and the total number of features when using the surrounding grids of the prediction point, as well as the number of features after feature selection based on correlation analysis.

	Train samples	Validation samples	Test samples	Original features	Dropped features
Ashimine	113 854	37 882	37 934	5702	239
Uchinoura	113 854	37 843	37 947	5702	296
Minamiaso	113 828	37 830	37 960	5702	502
Uwanokogen	108 642	37 570	37 947	5702	385
Imabari	113 802	37 830	37 934	5702	408
Sakai	113 776	37 830	37 947	5702	455
Kuwana	111 891	37 817	37 960	5702	354
Sekigahara	113 568	37 830	37 934	5702	437
Itoigawa	113 609	37 843	37 817	5702	516
Nobeyama	113 022	37 544	37 791	5702	869
Niijima	113 477	37 778	37 921	5702	236
Tsujido	113 763	37 791	37 960	5702	416
Saitama	113 789	37 817	37 960	5702	248
Kusatsu	112 359	37 271	37 752	5702	733
Kamaishi	113 464	37 687	37 921	5702	295
Oma	113 815	37 804	37 895	5702	256
Ishikari	113 243	37 791	37 934	5702	271
Kamikawa	113 607	36 907	37 791	5702	413

Period: 2019–2023 (1826 days)

Forecast cycles per day: 8 (3-hourly)

Lead times per cycle: 13 (3–39 h)

Locations: 18

Total rows: 3,418,272 (before removing missing observations)

2.2. LightGBM

Several previous studies on tabular data and post-processing model development have reported that decision tree-based models outperform other machine learning models in terms of training cost and performance scores. Among decision tree-based models, LightGBM (Ke et al., 2017) is particularly fast and high-performing (Iwase and Takenawa, 2024). Therefore, this study adopts LightGBM for the development of the post-processing model. LightGBM is an ensemble model based on decision trees, employing a boosting algorithm that sequentially adds decision trees and uses a quadratic optimization method.

Since LightGBM cannot handle multiple output variables simultaneously, a separate model should be created for each output variable. During training, Early Stopping was applied using validation data with a patience of 10 epochs, and the loss function was set to Mean Squared Error (MSE).

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{y}_i - y_i\right)^2 \tag{1}$$

where N is the number of data points, $\hat{y}_i$ is the predicted value, and y_i is the observed value.

2.3. Feature selection

When surrounding grids are included, the number of features reached 5702, which is excessive for LightGBM. Feature selection is necessary to reduce training time and prevent overfitting. To achieve this, the correlation analysis method proposed in Bedi and Toshniwal (2018), denoted as FS0 hereafter, was applied. FS0 computes pairwise absolute Pearson correlations among candidate features and removes redundancies.

To avoid removing critical input information, we exclude features at the grid closest to the prediction target and calendar variables from the analysis by dropping columns. Thus, the correlation analysis is performed solely among features from surrounding grids.

Let τ denote the correlation threshold. FS0 builds the absolute correlation matrix $[|r(x_i, x_j)|]_{i,j}$ and scans features in their column order. For each feature x_j, if there exists any earlier feature x_i $(i < j)$ with $|r(x_i, x_j)| > \tau$, then x_j is dropped (left-to-right rule). Based on validation data, the threshold was set to $\tau = 0.9$. The number of features after correlation analysis for each location is summarized in Table 3.

After applying the aforementioned feature selection process or in the case where only single-grid data were used, the following three further feature selection methods were applied to determine the optimal number of features:

- FS1. A method based on the correlation coefficient between each feature and the output variable.
- FS2. A method based on the mutual information between each feature and the output variable.
- FS3. A method based on feature importance calculated by LightGBM.

While many feature selection methods exist, this study adopted representative approaches (Chandrashekar and Sahin, 2014; Ross, 2014). The correlation coefficient-based method calculates the correlation coefficient between each feature and the output variable, and selects features in descending order of the correlation coefficient. The mutual information-based method uses mutual information between each feature and the output variable, where mutual information measures the degree of dependence between two variables (Ross, 2014). The LightGBM feature importance-based method calculates the contribution of each feature to reducing the loss function during tree splitting in LightGBM. Additionally, an analysis of the number of selected features was conducted. Variables such as prediction time, month, and hour were included in the final selected features regardless of the feature selection results.

2.4. Parameter tuning

Parameter tuning was conducted using Optuna (Akiba et al., 2019), a parameter optimization library based on Bayesian optimization algorithms. Specifically, this study employed LightGBMTuner (OPTUNA, 2024) to compute optimal values for each parameter iteratively. The tuning process was performed using a single representative location, and the tuned parameters were applied to other locations. The chosen representative location was Sekigahara, situated near the center of all target locations. Since LightGBM cannot handle multiple output variables, tuning was conducted separately for each output variable.

2.5. Neural network and CNN

To provide comparisons with neural-network-based approaches used in previous studies, we also implemented a feedforward neural network

表3 使用预测点周围网格时训练、验证和测试数据集中的样本数量，以及特征总数，以及基于相关分析进行特征选择后的特征数量。

	训练 样本	验证 样本	测试 样本 les	O原始 特 征	丢弃的 特征
Ashimin e	113 854	37 882	37 934	5702	239
Uchin oura	113 854	37 843	37 947	5702	296
Minami aso	113 828	37 830	37 960	5702	502
Uwanokoge n	108 64 2	37 570	37 947	5702	385
今治	113 80 2	37 830	37 934	5702	408
堺	113 776	37 830	37 947	5702	455
桑名	111 89 1	37 817	37 960	5702	354
Sekigahara I	113 568	37 830	37 934	5702	437
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ob新岛惠	113 0 22	37 544	37 791	5702	869
山	113 477	37 778	37 921	5702	236
Tsujido	113 763	37 791	37 960	5702	416
Saitama	113 789	37 817	37 960	5702	248
Kusatsu	112 359	37 271	37 752	5702	733
釜石市	113 464	37 687	37 921	5702	295
Oma	113 8 15	37 804	37 895	5702	256
石狩	113 243	37 791	37 934	5702	271
Kamikawa	113 607	36 907	37 791	5702	413

时期：2019–2023 （1826 天）

预报 周期 每日：8 （3小时一次）

每次循环的 预测 提前 时间：13 （3–39 小时）

地点：18

总 行数：3,418,272 （移除 缺失 观测值 前）

2.2. LightGBM

先前关于表格数据和后处理模型开发的几项研究表明，基于决策树的模型在训练成本和性能评分方面优于其他机器学习模型。在基于决策树的模型中，LightGBM（Ke等, 2017）尤其快速且高性能（Iwase and Takenawa, 2024）。因此，本研究采用LightGBM来开发后处理模型。LightGBM是一种基于决策树的集成模型，采用顺序添加决策树并使用二次优化方法的提升算法。

由于LightGBM无法同时处理多输出变量，因此需要为每个输出变量单独创建模型。训练过程中，使用验证数据应用早停，耐心值为10个周期，损失函数设置为均方误差(MSE)。

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(\hat{y}_i - y_i\right)^2 \tag{1}$$

其中 N 是数据点 的 数量， $\hat{y}_i$ 是 预测 值， 而 y_i 是 观测 值。

2.3. 特征 选择

当包含周围网格时，特征数量达到5702，这对LightGBM来说过于庞大。特征选择对于减少训练时间和防止过拟合是必要的。为此，采用了Bedi和Toshniwal (2018)中提出的相关分析方法（此后记为FS0）。FS0计算候选特征之间的成对绝对皮尔逊相关性，并去除冗余。

为 避免 删除 关键 输入 信息， 我们 通过 删除 列 将 最接近 预测 目标 的 网格 和 日历 变量 从 分析 中 排除 。 因此， 相关 分析 仅 在 来自 周围 网格 的 特征 之间 进行 。

让 τ 表示 相关 阈值。 FS0 构建 绝对 相关矩阵 $[|r(x_i, x_j)|]_{i,j}$ 并 按列 顺序 扫描特征 在其 列 中。

对于每个特征 x_j，如果存在任何较早的特征 x_i $(i < j)$ 使得 $|r(x_i, x_j)| > \tau$，则删除 x_j （从左到右规则）。基于验证数据，阈值设置为 $\tau = 0.9$。每个位置经过相关分析后的特征数量汇总于表3。

在 应用 上述 特征 选择 过程 后， 或 在 仅 使用 单网格 数据 的 情况下， 采用 了 以下 三种 进一步 的特征 选择 方法 来 确定 最优 特 征 数量：

FS1. 一种 基于 每个特征 与 输出变量 之间 相关系数 的 方法。 FS2. 一种 基于 每个特征 与 输出变量 之间 互信息 的 方法。 FS3. 一种 基于 LightGBM计算的特征重要性 的 方法。

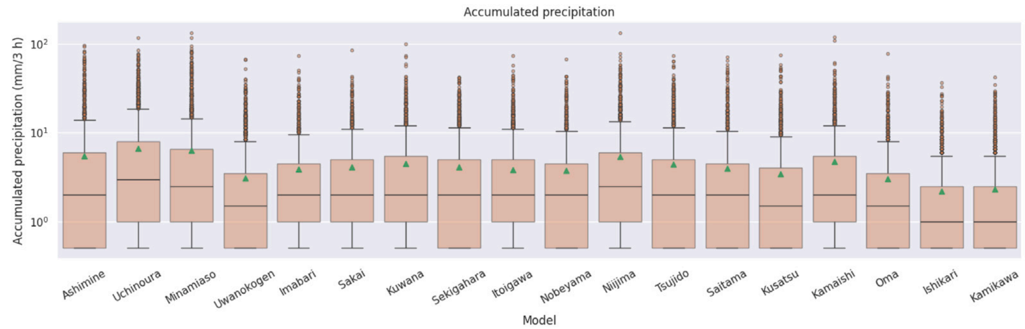
虽然存在许多特征选择方法，本研究采用了代表性方法（Chandrashekar and Sahin, 2014; Ross, 2014）。基于相关系数的方法计算每个特征与输出变量之间的相关系数，并按相关系数降序选择特征。基于互信息的方法使用每个特征与输出变量之间的互信息，其中互信息衡量两个变量之间的依赖程度（Ross, 2014）。基于LightGBM特征重要性的方法计算每个特征在LightGBM树分裂过程中对减少损失函数的贡献。此外，还对所选特征的数量进行了分析。无论特征选择结果如何，预测时间、月份和小时等变量始终包含在最终选择的特征中。

2.4. 参数 调优

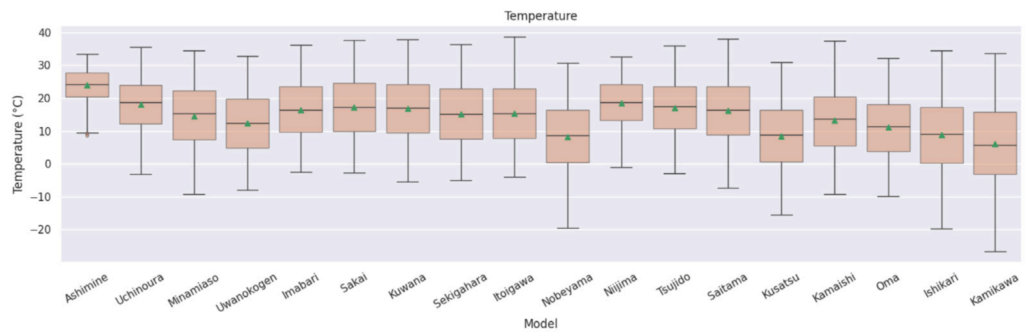
参数调优使用Optuna（Akiba et al., 2019）进行，Optuna是一个基于贝叶斯优化算法的参数优化库。具体而言，本研究采用LightGBMTuner（OPTUNA, 2024）迭代计算每个参数的最优值。调优过程使用单个代表性地点进行，调优后的参数应用于其他地点。选定的代表性地点是关原，位于所有目标地点的中心附近。由于LightGBM无法处理多输出变量，因此针对每个输出变量分别进行调优。

2.5. 神经网络 和 CNN

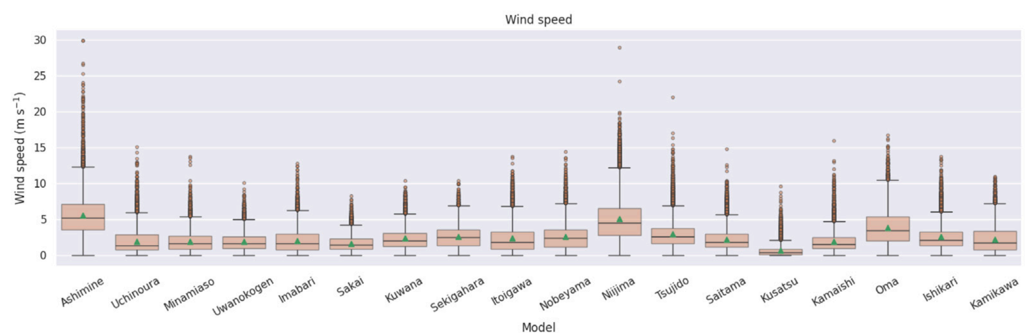
为了 与 先前 研究 中 使用 的 基于 神经网络 方法 进行 比较， 我们 还 实现 了一个 前馈 神经网络



(a) Precipitation



(b) Temperature



(c) Wind Speed

Fig. 2. Box plots of the distributions of each meteorological variable at each location. For precipitation, only nonzero values are shown (values > 0 mm) to avoid the dominance of zero precipitation and to provide an informative visualization; the precipitation axis is displayed on a logarithmic scale. The green triangle markers indicate the mean values.

and a CNN model. The structure of the neural network, implemented using TensorFlow (TensorFlow, 2024) with keras.layers, is as follows:

- Dense(1000, activation='relu')
- Dense(500, activation='relu')
- Dense(100, activation='relu')
- Dense(3)

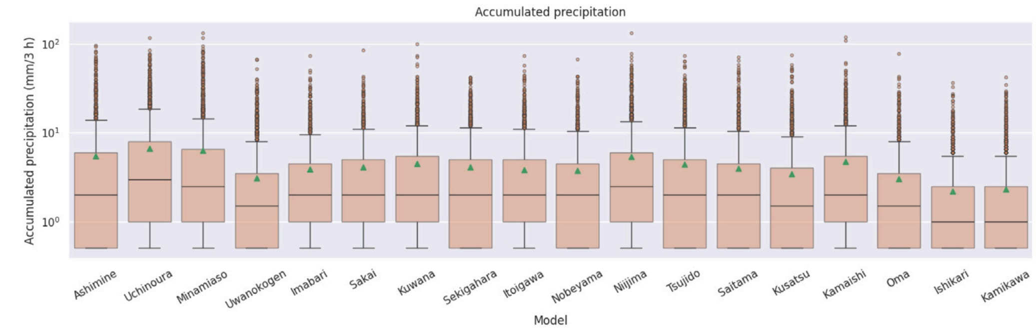
The input dimension corresponds to the number of features after FS0 selection, and the output dimension (3) corresponds to precipitation, temperature, and wind speed.

The model was trained with the Adam optimizer (learning rate 0.001) (Kingma, 2014), MSE as the loss, and Root Mean Squared Error (RMSE) :

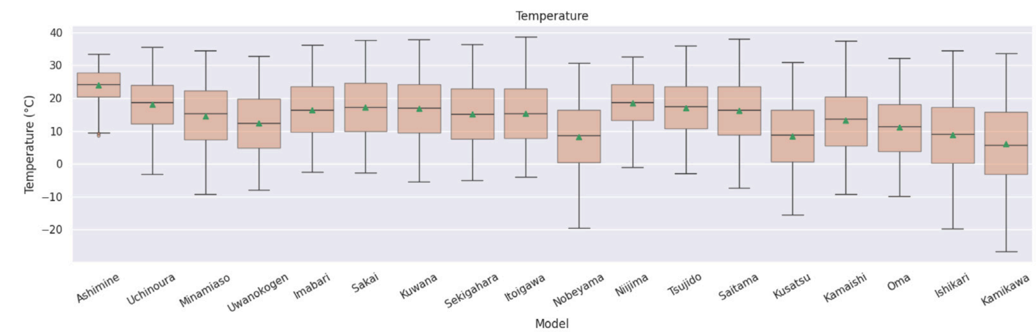
$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\hat{y}_i - y_i\right)^2} \tag{2}$$

as a metric. Training used early stopping with a patience of 10 epochs, based on validation RMSE. We set a maximum of 1000 epochs with a batch size of 64; in practice, training typically converged within about 25–50 epochs due to early stopping.

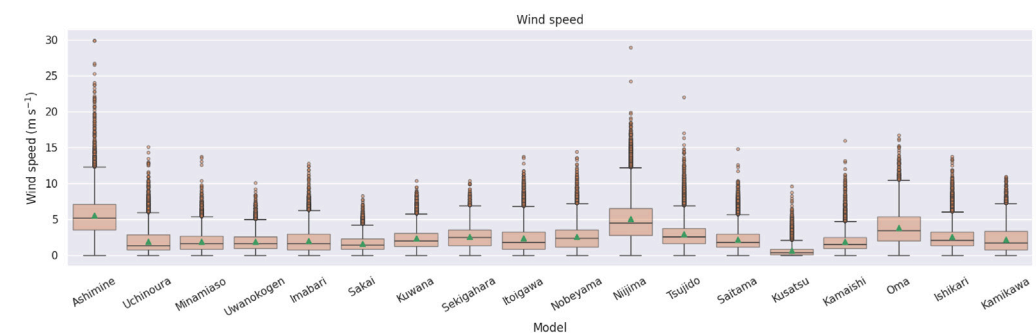
CNN models have many possible variations, including the number of layers, kernel sizes, numbers of channels, the use of skip connections, and the choice of optimization and regularization methods. A comprehensive comparison of these architectures is beyond the scope of the present paper. Accordingly, we limited the comparison to a CNN model based on prior work that used CNNs for post-processing MSM data.



(a) Precipitation



(b) Temperature



(c) Wind Speed

图 2. 各位置各气象变量分布的箱线图。对于降水，仅显示非零值（值> 0 mm），以避免零降水的主导地位并提供信息丰富的可视化；降水轴以对数尺度显示。绿色三角形标记表示平均值。

和一个 CNN 模型。 该 神经 网络 的 结构 ， 使用 TensorFlow 和 keras.layers (TensorFlow, 2024) 实现， 如下所示：

- 全连接层(1000, 激活=’ReLU’)
- 全连接层 (500, 激活 =’ReLU’)
- 全连接层 (100, 激活 =’ReLU’)
- 全连接层(3)

输入 维度 对应于 经过 FS0 选择后的 特征 数量， 输出 维度 （3） 对应于 降水、 温度和 风速。

该 模型 使用 Adam 优化器 （学习 率 0.001） （Kingma, 2014）进行 训练， 以 均方误差 为 损失函数， 并 以 均方根误差 （均方根误差）：

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\hat{y}_i - y_i\right)^2} \tag{2}$$

作为指标。 训练 使用了 早停法， 耐心 值 为 10 轮次， 基于 验证 均方根误差。 我们 设置了 最大 1000 轮次， 批量大小 为 64； 实际 上， 训练 通常 在 约 25–50 轮次 内 收敛， 由于 早停法。

CNN模型有许多可能的变体，包括层数、卷积核大小、通道数、是否使用跳跃连接，以及优化和正则化方法的选择。对这些架构进行全面比较超出了本文的范围。因此，我们将比较范围限定在先前的使用CNN对MSM数据进行后处理的工作基础上构建的CNN模型。

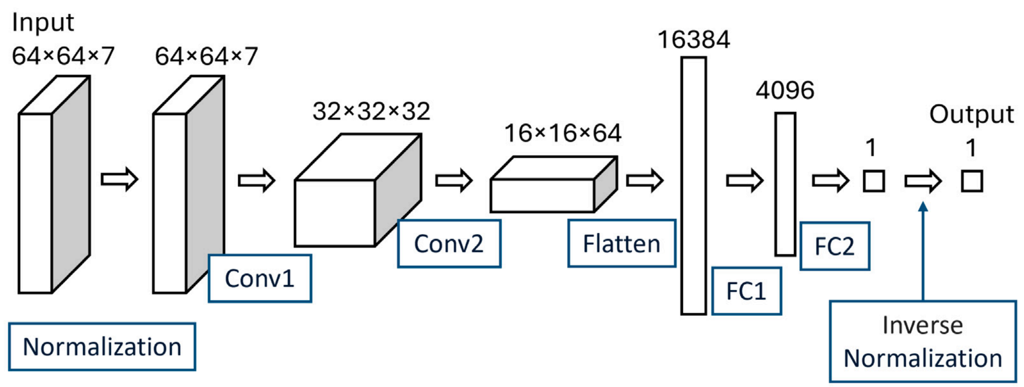


Fig. 3. Architecture of the CNN.

For the CNN baseline, we followed the experimental setting of Kudo (2022) to reproduce a representative prior CNN-based approach under comparable data conditions. Accordingly, the CNN was applied only to temperature prediction, as in the original study.

Our CNN uses a single forecast-time spatial field without a lookback window. The input is a $64 \times 64 \times 7$ tensor centered at each of the 18 locations. The seven channels consist of four surface-level fields (surface temperature, mean sea-level pressure, and surface wind components U and V) and temperature at three pressure levels (975, 925, and 850 hPa). Because the pressure-level grid is coarser than the surface grid, the pressure-level temperature fields are first cropped as 32×32 patches on the pressure-level grid and then upsampled to 64×64 to match the surface fields. The architecture of the CNN is shown in Fig. 3 and summarized in Table 4.

Normalization for CNN was performed as follows:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{3}$$

where x' represents the normalized value, x is the original value, and $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of the MSM input data. These minimum and maximum values were calculated for each prediction time (data sample) rather than across the entire training dataset. For temperature, $x_{\min}$ was adjusted to $x_{\min} - 3$ and $x_{\max}$ to $x_{\max} + 3$. An inverse normalization step was applied after the output layer, defined as:

$$y' = y \cdot (x_{\max} - x_{\min}) + x_{\min} \tag{4}$$

where y is the output from the output layer, and y' is the value after inverse normalization. These normalization and inverse-normalization procedures were introduced to reproduce the prior study as closely as possible, and should thus be understood as part of the baseline implementation rather than as independently optimized design choices in this study.

The learning settings for CNN, including the optimizer, loss function, and batch size, were the same as for NN. We set a maximum of 1000 epochs; in practice, training typically converged within about 15–20 epochs due to early stopping.

2.6. Evaluation metrics

In addition to RMSE, we used Mean Error (ME),

$$ME = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i),$$

to assess model bias, where N is the number of data points, $\hat{y}_i$ is the predicted value, and y_i is the observed value.

In addition, the following metrics were used to further assess model performance for specific variables.

Table 4
Summary of processing at each CNN layer. Each layer's implementation is described using TensorFlow (TensorFlow, 2024) keras.layers.

Layer	Function
Conv1	ZeroPadding2D(padding=(2, 2)) Conv2D(32, kernel_size=5, strides=1, padding='valid') MaxPooling2D(pool_size=2, strides=2) BatchNormalization() Activation('relu')
Conv2	ZeroPadding2D(padding=(2, 2)) Conv2D(64, kernel_size=5, strides=1, padding='valid') MaxPooling2D(pool_size=2, strides=2) BatchNormalization() Activation('relu')
FC1	Dense(4096) BatchNormalization() Activation('relu')
FC2	Dense(1)

2.6.1. Statistical tests for temperature and wind speed

Statistical comparisons were conducted using paired absolute errors. Let

$$e_i^{\text{main}} = \left|\hat{y}_i^{\text{main}} - y_i\right|, \qquad e_i^{\text{base}} = \left|\hat{y}_i^{\text{base}} - y_i\right|$$

denote the absolute errors of the main and baseline models for sample i. Then the mean absolute errors (MAEs) are defined as

$$\text{MAE}_{\text{main}} = \frac{1}{N}\sum_{i=1}^{N} e_i^{\text{main}}, \qquad \text{MAE}_{\text{base}} = \frac{1}{N}\sum_{i=1}^{N} e_i^{\text{base}},$$

and the paired differences are defined by

$$d_i = e_i^{\text{main}} - e_i^{\text{base}}.$$

Based on these paired differences, we conducted a paired t-test and a Wilcoxon signed-rank test for temperature and wind speed. Because of the very large sample size, these tests are interpreted only as supplementary information, and practical significance is assessed mainly from the error metrics themselves.

2.6.2. Event-based metrics for precipitation

For precipitation, forecast skill was additionally evaluated using event-based metrics. For each precipitation threshold τ, forecasts and observations were converted into binary events. Let $H(\tau)$, $M(\tau)$, and $F(\tau)$ denote the numbers of hits, misses, and false alarms, respectively,

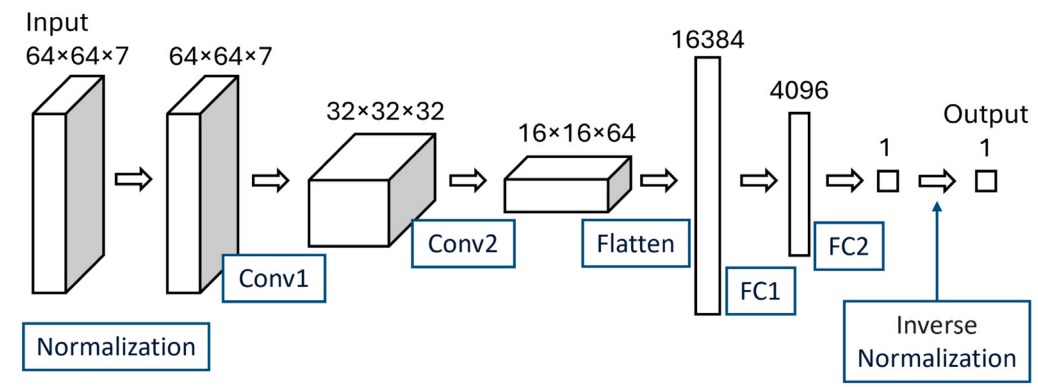


图 3. CNN的架构 。

对于CNN基线，我们遵循Kudo（2022年）的实验设置，在可比的数据条件下复现了一个具有代表性的基于CNN的先前方法。因此，与原始研究一样，CNN仅应用于温度预测。

我们的CNN使用单个预报时间空间场，没有回溯窗口。输入是一个以18个地点每个为中心的张量$64 \times 64 \times 7$。七个通道包括四个地表场（地表温度、平均海平面气压和地表风速分量U 和V）以及三个气压层（975、925和850百帕）的温度。由于气压层网格比地表网格更粗糙，气压层温度场首先在气压层网格上裁剪为32×32 的块，然后上采样到 64×64 以匹配地表场。CNN的架构如图3所示，并在表4中总结。

归一化 对于 CNN 按照 以下 方式 进行:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{3}$$

其中 x' 表示归一化值，x 是原始值，$x_{\min}$ 和$x_{\max}$ 是MSM输入数据的最小值和最大值。这些最小值和最大值是针对每个预测时间（数据样本）计算的，而不是在整个训练数据集上计算。对于温度，$x_{\min}$ 调整为 $x_{\min} - 3$ ，$x_{\max}$ 调整为$x_{\max} + 3$。在输出层之后应用逆归一化步骤，定义如下：

$$y' = y \cdot (x_{\max} - x_{\min}) + x_{\min} \tag{4}$$

其中 y 是输出层的输出，y' 是逆归一化后的值。引入这些归一化和逆归一化过程是为了尽可能复现先前的研究，因此应将其视为基线实现的一部分，而非本研究中独立优化的设计选择。

CNN 的 学习 设置 ， 包括 优化器 、 损失 函数 和 批量 大小 ， 与 NN 相同 。 我们 设置了 最大 1000 个 轮次 ； 在 实践中 ， 训练 通常 会在 约 15–20 个 轮次 内 收敛 ， 得益于早停。

2.6. 评估 指标

除 RMSE 外， 我们 还使用了 平均误差 （ ME ），

$$ME = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i),$$

用于 评估 模型 偏差 ， 其中 N 是 数据 点 数量 ， $\hat{y}_i$ 是 预测值 ， 而 y_i 是 观测值。

此 外 ， 还 使 用 了 以 下 指 标 来 进 一 步评估特定变量的模型性能。

表 4 摘要 ： 每个 CNN 层 的 处理过程。 每个 层的 实现 使用 TensorFlow （ TensorFlow ， 2024） keras.layers 进行描述。

层	功能
Conv1	ZeroPadding2D(padding=(2, 2)) 积核大小=5, 步长=1, 填充='valid') 2D最大池化(pool_size=2, 步长=2) Batch归一化() 激活('ReLU')
Conv2	添加2D（填充=(2, 2)) ZeroP 2D卷积(64 步长 =1, 填充 =' valid') 2D最大池化(池大小=2, 步长=2) 批 激活(' relu')
FC1	全连接层(4096) 批量归一化() 激活('relu')
FC2	Dense(1)

2.6.1. 温度 和 风速 的 统计 检验

通 过 配 对 绝 对 误 差进行了统计比较。让

$$e_i^{\text{main}} = \left|\hat{y}_i^{\text{main}} - y_i\right|, \qquad e_i^{\text{base}} = \left|\hat{y}_i^{\text{base}} - y_i\right|$$

用 i 表示 主模型 和 基线模型 的 绝对误差 。 然后 定义 平均绝对误差 (平均绝对误差) 为

$$\text{MAE}_{\text{main}} = \frac{1}{N}\sum_{i=1}^{N} e_i^{\text{main}}, \qquad \text{MAE}_{\text{base}} = \frac{1}{N}\sum_{i=1}^{N} e_i^{\text{base}},$$

并 将 配对差异 定义为

$$d_i = e_i^{\text{main}} - e_i^{\text{base}}.$$

d

基于这些配对差异，我们对温度和风速进行了配对t-检验和威尔科克森符号秩检验。由于样本量非常大，这些检验仅作为补充信息解释，实际意义主要从误差指标本身评估。

2.6.2. 基于事件 的降水 指标

对于 降水， 预报 技巧 还 额外 使用 基于事件 的 指标 进行评估。 对于 每个 降水 阈值 τ， 预报 和 观测值 被 转换为 二元 事件。 让 $H(\tau)$、 $M(\tau)$ 和 $F(\tau)$ 分别表示 命中 、 漏报 和 虚警 的数量 ，

at threshold τ. Then, the event-based metrics were defined as

$$\mathrm{TS}(\tau) = \frac{H(\tau)}{H(\tau) + M(\tau) + F(\tau)},$$
$$\mathrm{POD}(\tau) = \frac{H(\tau)}{H(\tau) + M(\tau)},$$
$$\mathrm{FAR}(\tau) = \frac{F(\tau)}{H(\tau) + F(\tau)},$$
$$\mathrm{Bias}(\tau) = \frac{H(\tau) + F(\tau)}{H(\tau) + M(\tau)}.$$

2.7. Weighted tweedie model for precipitation

Precipitation is a nonnegative variable with a highly skewed distribution and many zero or near-zero values. Under such conditions, standard regression training based on mean squared error tends to be dominated by the large number of weak-rainfall or no-rainfall samples, and may therefore lead to predictions biased toward intermediate values, without sufficiently improving forecast accuracy for moderate to heavy precipitation events. To address these characteristics, in this study we examined a model that uses a Tweedie loss function with sample weight coefficients in LightGBM training for precipitation prediction (hereafter referred to as the weighted Tweedie model) (Jørgensen, 1997; Ke et al., 2017; Dunn and Smyth, 2005). In preliminary experiments for a few sites, the weighted Tweedie model showed more improvements than the variants using only weighting or only the Tweedie loss, so we focus on this variant here.

Let $y_i \geq 0$ denote the observed precipitation for sample i, and let $\mu_i > 0$ denote the predicted mean precipitation. The Tweedie distribution belongs to the exponential dispersion family and satisfies (Jørgensen, 1997; Dunn and Smyth, 2005)

$$\mathrm{Var}(Y_i) = \phi \mu_i^p,$$

where $\phi > 0$ is the dispersion parameter and p is the Tweedie variance power. In this study, the exponent p was treated as a tunable hyperparameter.

The sample weights were defined as a monotone nondecreasing piecewise linear function of the observed precipitation. Let the knot positions be

$$x_0 = 0, \quad x_1 = 5, \quad x_2 = 10, \quad x_3 = 15, \quad x_4 = 20 \text{ mm},$$

and the corresponding weight values be

$$w_0,\ w_1,\ w_2,\ w_3,\ w_4.$$

A monotonicity constraint,

$$1 = w_0 \leq w_1 \leq w_2 \leq w_3 \leq w_4,$$

was imposed. Then, the sample weight function $w(y)$ was defined by linear interpolation between adjacent knots as follows:

$$w(y) = \begin{cases} w_0 + \dfrac{w_1 - w_0}{x_1 - x_0}(y - x_0), & x_0 \leq y \leq x_1, \\ w_1 + \dfrac{w_2 - w_1}{x_2 - x_1}(y - x_1), & x_1 < y \leq x_2, \\ w_2 + \dfrac{w_3 - w_2}{x_3 - x_2}(y - x_2), & x_2 < y \leq x_3, \\ w_3 + \dfrac{w_4 - w_3}{x_4 - x_3}(y - x_3), & x_3 < y. \end{cases}$$

In the implementation, the resulting weights were clipped to the range

$$1 \leq w(y) \leq w_{\max},$$

where $w_{\max} = 10$. Thus, samples with higher precipitation intensity contributed more strongly to training, while the monotonicity constraint maintained the stability and interpretability of the weighting scheme.

Using these weights, the precipitation model was trained by minimizing the weighted Tweedie loss

$$\mathcal{L}_{\mathrm{train}} = \sum_{i=1}^{N} w(y_i)\, \ell_{\mathrm{Tw}}(y_i, \mu_i),$$

where $\ell_{\mathrm{Tw}}(y_i, \mu_i)$ denotes the Tweedie loss for sample i. In this study, we used the Tweedie objective implemented in LightGBM (Ke et al., 2017; LightGBM Contributors, 2026). Except for the precipitation loss function and sample-weighting scheme described below, the input data configuration and the tree-related LightGBM settings were the same as those of the "around all tune" model.

Although the model was trained using the weighted Tweedie loss, the hyperparameters were selected in Optuna based on an event-based evaluation score, because the purpose of weighting was to improve performance at relevant precipitation thresholds. The tuned hyperparameters were the Tweedie variance power p and the parameters determining the shape of the weight function, and the tuning was carried out separately for each observation site because the behavior of the weighted Tweedie model showed strong site dependence. A monotone nondecreasing constraint was imposed on the weight function so that the weight would not decrease as precipitation intensity increased.

Using the event-based metrics defined in Section 2.6.2, the TS values across thresholds were aggregated using a geometric mean:

$$S_{\mathrm{geo}} = \exp\left(\frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \log\left(\max(\mathrm{TS}(\tau), \varepsilon)\right) \right).$$

Furthermore, to suppress overprediction or underprediction of event frequency, a penalty was imposed on the forecast Bias at selected thresholds. Using this, the total score employed in Optuna was defined as

$$S_{\mathrm{total}} = S_{\mathrm{geo}} - \sum_{\tau \in \mathcal{T}_{\mathrm{bias}}} \gamma_\tau \left| \log\left(\max(\mathrm{Bias}(\tau), b_{\min})\right) \right|.$$

Here, $\gamma_\tau \geq 0$ denotes the penalty coefficient, $\mathcal{T}_{\mathrm{bias}}$ denotes the set of thresholds to which the bias penalty is applied, and $b_{\min}$ is a small positive constant for numerical stability. This objective function was intended to encourage models with balanced detection skill across multiple thresholds while reducing substantial overprediction or underprediction of event frequency.

In the present implementation, the thresholds used for optimization were set to

$$\mathcal{T} = \{1, 2, \ldots, 20\} \text{ mm},$$

and equal weights were used for all thresholds in the geometric mean. In addition, the bias penalty was applied at 1, 5, 10, and 15 mm with $\gamma_\tau = 0.05$ for all of these thresholds, and $\varepsilon = 10^{-6}$ and $b_{\min} = 10^{-6}$ were set for numerical stability. The resulting model is referred to as the weighted Tweedie model in the following sections.

3. Results and discussion

3.1. Model comparison

3.1.1. RMSE and ME

First, Tables 5 and 6 present the RMSE and ME of each model, as well as for MSM and MSMG, on the validation and test datasets.

For RMSE, both validation and test datasets show that the models using LightGBM outperform MSM and MSMG. Similarly, the neural network and CNN models perform worse compared to those using LightGBM. Moreover, models incorporating data from surrounding grids (around) exhibit slightly better performance than those using only the nearest grid (1grid). This indicates that leveraging features from surrounding grids and reducing them through feature selection allows for the inclusion of informative data that improves accuracy. The correlation analysis-based feature selection (FS0) of surrounding grid data yielded the best results in most cases. Among the three feature selection methods (FS1 - FS3) described in Section 2.3, no performance improvement was observed. For the number of features in FS1 to FS3, we used the values that performed best for each feature selection method and each output variable in Section 3.2.

在 阈值 τ 处。 然后， 基于事件 的 指标 被定义为

$$\mathrm{TS}(\tau) = \frac{H(\tau)}{H(\tau) + M(\tau) + F(\tau)},$$
$$\mathrm{POD}(\tau) = \frac{H(\tau)}{H(\tau) + M(\tau)},$$
$$\mathrm{FAR}(\tau) = \frac{F(\tau)}{H(\tau) + F(\tau)},$$
$$\mathrm{Bias}(\tau) = \frac{H(\tau) + F(\tau)}{H(\tau) + M(\tau)}.$$

2.7. 加权 *Tweedie* 模型 用于 降水

降水是一个非负变量，具有高度偏斜的分布和许多零或接近零的值。在这种条件下，基于均方误差的标准回归训练往往被大量弱降水或无降水样本所主导，因此可能导致预测偏向于中间值，而无法充分提高中度到强降水事件的预报精度。为了解决这些特征，在本研究中，我们检验了一种在LightGBM训练中使用Tweedie损失函数和样本权重系数进行降水预测的模型（以下简称加权Tweedie模型）（Jørgensen, 1997; Ke等, 2017; Dunn和Smyth, 2005）。在少数站点的初步实验中，加权Tweedie模型比仅使用加权或仅使用Tweedie损失的变体显示出更多的改进，因此我们在此重点关注这种变体。

令 $y_i \geq 0$ 表示样本 i的观测降水，并令 $\mu_i > 0$ 表示预测的平均降水。Tweedie分布属于指数分散族，并满足（Jørgensen, 1997; Dunn和Smyth, 2005）

$$\mathrm{Var}(Y_i) = \phi \mu_i^p,$$

其中 $\phi > 0$ 是 分散 参数 而 p 是 Tweedie 方差 幂。 在 本 研究中, 指数 p 被 视为 可调超参数。

样本 权重 被定义为 观测 降水 的 单调非减 分段线性函数。 令 节点 位置为

$$x_0 = 0, \quad x_1 = 5, \quad x_2 = 10, \quad x_3 = 15, \quad x_4 = 20 \text{ mm},$$

且 对应 的 权重 值为

$$w_0,\ w_1,\ w_2,\ w_3,\ w_4.$$

施加了单调性约束,

$$1 = w_0 \leq w_1 \leq w_2 \leq w_3 \leq w_4,$$

已 施加。 然后， 通过 相邻 节点 之间的 线性插值，将样本权重函数 $w(y)$ 定义 如下:

$$w(y) = \begin{cases} w_0 + \dfrac{w_1 - w_0}{x_1 - x_0}(y - x_0), & x_0 \leq y \leq x_1, \\ w_1 + \dfrac{w_2 - w_1}{x_2 - x_1}(y - x_1), & x_1 < y \leq x_2, \\ w_2 + \dfrac{w_3 - w_2}{x_3 - x_2}(y - x_2), & x_2 < y \leq x_3, \\ w_3 + \dfrac{w_4 - w_3}{x_4 - x_3}(y - x_3), & x_3 < y. \end{cases}$$

在实现中，生成的权重被 裁剪到 范围

$$1 \leq w(y) \leq w_{\max},$$

其中 $w_{\max} = 10$。 因此， 降水强度 较高 的 样本 对 训练 贡献 更大 , 而 单调性约束 保持 了 加权方案 的 稳定性和 可解释性 。

使用 这些 权重， 通过 最小化 加权 Tweedie 损失 来 训练 降水 模型 。

$$\mathcal{L}_{\mathrm{train}} = \sum_{i=1}^{N} w(y_i)\, \ell_{\mathrm{Tw}}(y_i, \mu_i),$$

其中 $\ell_{\mathrm{Tw}}(y_i, \mu_i)$ 表示样本 i 的 Tweedie 损失。本研究中，我们使用了LightGBM（Ke等, 2017；LightGBM贡献者, 2026）实现的 Tweedie 目标函数。除下文描述的降水损失函数和样本加权方案外，输入数据配置和树相关 LightGBM 设置与“around all tune”模型保持一致。

虽然模型使用加权Tweedie损失进行训练，但超参数在Optuna中基于基于事件的评估分数进行选择，因为加权的目的是提高相关降水阈值下的性能。调优的超参数是Tweedie方差幂 p 以及决定权重函数形状的参数，并且由于加权Tweedie模型的行为表现出强烈的站点依赖性，因此对每个观测站点分别进行调优。对权重函数施加了单调非递减约束，以确保权重不会随着降水强度的增加而减少。

使用 第 2.6.2 节 定义的 基于事件的 指标 ， 各阈值 的 TS 值 通过 几何平均 进行聚合:

$$S_{\mathrm{geo}} = \exp\left(\frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \log\left(\max(\mathrm{TS}(\tau), \varepsilon)\right) \right).$$

此外，为了抑制事件频率的 高估 或 低估 ， 对选定阈值 的 预报 偏差 施加了 惩罚 。 通过这种方式， Optuna 中使用的 总得分 定义为

$$S_{\mathrm{total}} = S_{\mathrm{geo}} - \sum_{\tau \in \mathcal{T}_{\mathrm{bias}}} \gamma_\tau \left| \log\left(\max(\mathrm{Bias}(\tau), b_{\min})\right) \right|.$$

这里，$\gamma_\tau \geq 0$ 表示惩罚系数，$\mathcal{T}_{\mathrm{bias}}$ 表示应用偏差惩罚的阈值集合，$b_{\min}$ 是为数值稳定性设置的一个小正常数。该目标函数旨在鼓励模型在多个阈值上具有平衡的检测技能，同时减少对事件频率的显著过度预测或欠预测。

在 当前 实现中， 用于 优化的 阈值 设置 为

$$\mathcal{T} = \{1, 2, \ldots, 20\} \text{ mm},$$

并且在几何平均中，所有阈值使用相等的权重。此外，偏差惩罚应用于1、5、10和15毫米，对这些阈值均采用$\gamma_\tau = 0.05$ ，并设置$\varepsilon = 10^{-6}$ 和 $b_{\min} = 10^{-6}$ 以保证数值稳定性。得到的模型在后续章节中称为加权Tweedie模型。

3. 结果 与 讨论

3.1. 模型 比较

3.1.1. 均方根误差 与 平均误差

首先， 表 5 和6 展示了 各 模型的 均方根误差 和 平均误差 ， 以及 MSM 和 MSMG 在 验证 和 测试 数据集上的 结果。

对于均方根误差，验证和测试数据集均显示使用LightGBM的模型优于MSM和MSMG。同样，神经网络和CNN模型的性能逊于使用LightGBM的模型。此外，结合周围网格数据（周围）的模型性能略优于仅使用最近网格（单网格模型）的模型。这表明利用周围网格的特征并通过特征选择进行缩减，可以纳入改善准确性的信息数据。基于相关分析的特征选择（FS0）对周围网格数据在大多数情况下取得了最佳结果。在第2.3节所述的三种特征选择方法（FS1-FS3）中，未观察到性能提升。对于FS1到FS3的特征数量，我们使用了在第3.2节中对每种特征选择方法和每个输出变量表现最佳的值。

Table 5
RMSE and ME for validation data. "1grid" represents models using data from only one grid, "around" indicates the use of surrounding grid data, "nn" denotes neural network, "all" refers to data before three feature selection methods (FS1 - FS3), "pcc" indicates the correlation coefficient-based method (FS1), "mi" denotes the mutual information method (FS2), "gbm" represents the feature importance method by LightGBM (FS3), and "tune" refers to parameter tuning. Averages are calculated across prediction times and locations.

	RMSE			ME		
	Precip.	Temp.	Wind	Precip.	Temp.	Wind
MSM	2.948	1.836	2.096	0.048	−0.405	0.753
1grid nn	2.555	1.472	1.167	−0.075	0.130	−0.013
1grid all	2.510	1.364	1.070	−0.007	0.084	−0.018
1grid pcc	2.526	1.367	1.070	−0.003	0.085	−0.018
1grid mi	2.535	1.420	1.070	**−0.001**	0.096	**−0.016**
1grid gbm	2.518	1.363	1.073	−0.006	0.085	−0.018
around all	2.501	1.332	1.050	**0.001**	0.081	−0.025
around pcc	2.496	1.344	1.053	0.006	0.082	−0.026
around mi	2.505	1.342	1.051	0.004	0.083	−0.026
around gbm	2.502	1.333	1.051	−0.002	0.080	−0.027
around all tune	**2.489**	**1.326**	**1.046**	0.009	**0.074**	−0.023
CNN		1.487			0.086	

Table 6
RMSE and ME for test data.

	RMSE			ME		
	Precip.	Temp.	Wind	Precip.	Temp.	Wind
MSM	2.759	1.811	1.999	0.085	−0.340	0.673
MSMG	2.726	1.394	1.237	0.107	**0.014**	0.159
1grid nn	2.439	1.485	1.175	−0.064	0.143	−0.060
1grid all	2.387	1.368	1.075	0.020	0.110	−0.058
1grid pcc	2.390	1.369	1.076	**0.016**	0.109	−0.058
1grid mi	2.409	1.428	1.075	0.028	0.117	**−0.057**
1grid gbm	2.392	1.368	1.078	0.017	0.111	−0.058
around all	2.358	1.341	1.053	0.031	0.111	−0.062
around pcc	2.374	1.349	1.056	0.030	0.124	−0.062
around mi	2.381	1.347	1.055	0.033	0.119	−0.061
around gbm	2.361	1.341	**1.052**	0.028	0.112	−0.063
around all tune	**2.344**	**1.335**	**1.052**	0.033	0.106	−0.059
CNN		1.501			0.179	

For ME, models using LightGBM tended to reduce bias in both validation and test datasets. However, for precipitation in the validation dataset, MSM achieved the best ME, while for temperature in the test dataset, MSMG performed the best.

For subsequent predictions in this study, we used the "around all tune" model, which showed the best overall performance; here, "around" indicates the use of surrounding-grid predictors, "all" indicates that all selected predictors were retained without further reduction, and "tune" indicates hyperparameter optimization.

Fig. 4 shows the prediction results for test data compared with MSM, MSMG, and observed values at selected locations and prediction times. In the case of 3-hour precipitation at Kamikawa and 18-hour wind speed at Itoigawa, post-processing models corrected the apparent large bias. On the other hand, for 18-hour temperature at Itoigawa, partial corrections by the post-processing model were observed.

3.1.2. Statistical tests and event-based metrics

For temperature and wind speed, Table 7 summarizes the paired t-test and Wilcoxon signed-rank test results comparing the main model (around all tune) with the baseline models (MSM and MSMG), based on the metrics defined in Section 2.6.1. The table reports $\mathrm{MAE}_{\mathrm{main}}$, $\mathrm{MAE}_{\mathrm{base}}$, the standard deviation of the paired differences, and the corresponding one-sided p-values. Because of the very large sample size, these test results should be interpreted only as supplementary information, and practical significance should be assessed mainly from the error metrics themselves.

For precipitation, Table 8 summarizes the event-based metrics defined in Section 2.6.2—TS, POD, FAR, and Bias—for four precipitation thresholds. Here, "around all tune" denotes our original LightGBM-based post-processing model, and "weighted Tweedie" denotes the weighted Tweedie model.

Table 7
Paired t-test and Wilcoxon signed-rank test results comparing the LightGBM "around all tune" model (main model) with the baseline models for temperature and wind speed. The table reports the mean absolute errors (MAEs) of the main and baseline models, the standard deviation of the paired differences in absolute error ($\mathrm{SD}_{\mathrm{diff}}$), and the corresponding one-sided p-values. All tests were performed over $N = 682,305$ paired samples.

Variable	Baseline	$\mathrm{MAE}_{\mathrm{main}}$	$\mathrm{MAE}_{\mathrm{base}}$	$\mathrm{SD}_{\mathrm{diff}}$	p_t	p_W
Temperature	MSM	0.9888	1.3964	1.0840	0.000	0.000
Temperature	MSMG	0.9888	1.0275	0.7675	0.000	0.000
Wind Speed	MSM	0.7549	1.4186	1.3530	0.000	0.000
Wind Speed	MSMG	0.7549	0.9035	0.6980	0.000	0.000

Table 8
Event-based verification metrics for precipitation, pooled over all locations and forecast lead times. Higher TS and POD indicate better detection skill, lower FAR indicates fewer false alarms, and Bias values closer to 1 indicate better frequency matching. Here, "around all tune" denotes the original LightGBM-based post-processing model.

Threshold (mm)	Model	TS	POD	FAR	Bias
1.0	MSM	0.4311	0.6373	0.4287	1.1156
1.0	MSMG	**0.4555**	0.6558	**0.4014**	**1.0956**
1.0	around all tune	0.4293	0.6931	0.4699	1.3075
1.0	weighted Tweedie	0.4435	**0.7005**	0.4527	1.2800
5.0	MSM	0.3226	0.5061	0.5293	1.0753
5.0	MSMG	**0.3344**	0.5014	0.4991	**1.0009**
5.0	around all tune	0.3059	0.3943	**0.4228**	0.6831
5.0	weighted Tweedie	0.3316	**0.5208**	0.5228	1.0912
10.0	MSM	0.2450	0.4084	0.6201	1.0749
10.0	MSMG	0.2485	**0.4122**	0.6152	1.0713
10.0	around all tune	0.1966	0.2356	**0.4571**	0.4340
10.0	weighted Tweedie	**0.2520**	0.3979	0.5927	**0.9771**
15.0	MSM	0.1902	0.3189	0.6796	**0.9956**
15.0	MSMG	**0.2096**	**0.3551**	0.6616	1.0492
15.0	around all tune	0.1251	0.1420	**0.4877**	0.2772
15.0	weighted Tweedie	0.1883	0.2719	0.6201	0.7158

Overall, MSMG showed favorable performance among the compared models. The "around all tune" model achieved the highest POD at the 1.0 mm threshold, indicating high sensitivity to light precipitation events. However, its FAR and Bias were also the largest, suggesting that it tended to overpredict precipitation occurrence. At thresholds of 5.0 mm and above, its TS and POD became lower than those of MSM and MSMG, and its Bias fell well below 1 at 10.0 mm and 15.0 mm, indicating underdetection of heavier precipitation events. Compared with this behavior, the weighted Tweedie model improved TS and Bias at 10.0 mm and increased POD at 5.0 mm. Although its overall performance was still slightly below that of MSMG, it showed a better balance across the precipitation thresholds.

3.2. Comparison of feature selection methods

Here, we conducted experiments at a single location in Sekigahara, similar to parameter tuning, to evaluate the influence of the number of selected features in the three feature selection methods (FS1 - FS3) proposed in Section 2.3. For both cases where only the data of the grid closest to the prediction point (1grid) and the data including surrounding grids (around) were used, the three feature selection methods were applied, and models were constructed by incrementally increasing the number of selected features k by 10. Fig. 5 summarizes the RMSE on the validation data for each feature count k. The results indicate that the accuracy tends to improve as the number of features increases, but stabilizes when the number of features exceeds a certain threshold. This suggests that further feature selection after correlation analysis does not enhance accuracy.

表5 验证数据的均方根误差和平均误差。"单网格模型"表示仅使用一个网格数据的模型，"周围网格数据"表示使用周围网格数据，"神经网络"表示神经网络，"全部数据"表示三种特征选择方法（FS1 - FS3）之前的数据，"皮尔逊相关系数法"表示基于相关系数的方法（FS1），"互信息方法"表示互信息方法（FS2），"特征重要性"表示LightGBM的特征重要性方法（FS3），"参数调优"表示参数调优。平均值在预测时间和地点上计算。

	RMSE			ME		
	降水	气温	Wind	降水	气温	Wind
MSM	2.948	1.836	2.096	0.048	−0.405	0.753
单网格模型 n n	2.555	1.472	1.167	−0.075	0.130	−0.013
1grid all	2.510	1.364	1.070	−0.007	0.084	−0.018
单网格模型 皮尔逊相关系数法	2.526	1.367	1.070	−0.003	0.085	−0.018
单网格模型 互信息法	2.535	1.420	1.070	**−0.001**	0.096	**−0.016**
1网格 gb m	2.518	1.363	1.073	−0.006	0.085	−0.018
周围网格数据 a ll	2.501	1.332	1.050	**0.001**	0.081	−0.025
around 皮尔逊相关系数法	2.496	1.344	1.053	0.006	0.082	−0.026
周围网格数据 互信息法	2.505	1.342	1.051	0.004	0.083	−0.026
周围网格数据 梯度提升模型	2.502	1.333	1.051	−0.002	0.080	−0.027
周围网格数据 全部数据 参数调优	**2.489**	**1.326**	**1.046**	0.009	**0.074**	−0.023
CNN		1.487			0.086	

表 6 均方根误差 和 平均误差 对 测试 数据。

	RMSE			ME		
	降水	气温	Wind	降水	气温	Wind
MSM	2.759	1.811	1.999	0.085	−0.340	0.673
MSMG	2.726	1.394	1.237	0.107	**0.014**	0.159
单网格模型 神经网络	2.439	1.485	1.175	−0.064	0.143	−0.060
单网格模型 全部数据	2.387	1.368	1.075	0.020	0.110	−0.058
单网格模型 皮尔逊相关系数法	2.390	1.369	1.076	**0.016**	0.109	−0.058
1grid mi	2.409	1.428	1.075	0.028	0.117	**−0.057**
1网格 gb m	2.392	1.368	1.078	0.017	0.111	−0.058
周围网格数据 a ll	2.358	1.341	1.053	0.031	0.111	−0.062
around 皮尔逊相关系数法	2.374	1.349	1.056	0.030	0.124	−0.062
周围 互信息法	2.381	1.347	1.055	0.033	0.119	−0.061
around gbm	2.361	1.341	**1.052**	0.028	0.112	−0.063
周围 全部数据 参数调优	**2.344**	**1.335**	**1.052**	0.033	0.106	−0.059
CNN		1.501			0.179	

对于 平均误差， 使用 LightGBM 的 模型 倾向于 在 验证 和 测试 数据集 中 减少 偏差。 然而， 对于 验证 数据集 中的 降水， MSM 获得了 最佳 平均误差， 而对于 测试 数据集 中的 温度， MSMG 表现 最佳。

在本研究的后续预测中，我们使用了"around all tune"模型，该模型展现了最佳的整体性能；其中，"around"表示使用周围网格预测因子，"all"表示保留所有选定预测因子而不进一步减少，"tune"表示超参数优化。

图4显示了测试数据的预测结果与MSM、MSMG以及选定位置和预测时间观测值的对比。在Kamikawa的3小时降水和Itoigawa的18小时风速案例中，后处理模型修正了明显的大偏差。另一方面，对于Itoigawa的18小时温度，观察到了后处理模型的部分修正。

3.1.2. 统计 检验 与 基于事件的 指标

对于温度和风速，表7总结了配对t检验和威尔科克森符号秩检验的结果，这些结果比较了主模型（around all tune）与基线模型（MSM和MSMG），基于第 2 6 1 节定义的指标。$\mathrm{MAE}_{\mathrm{main}}$ $\mathrm{MAE}_{\mathrm{base}}$ 表格报告了配对差异标准差以及相应的单侧p值。由于样本量非常大，这些检验结果只能作为补充信息来解释，实际意义主要应从误差指标本身来评估。

对于 降水， 表 8 总结 了 基于事件 的指标 ， 这些指标定义 于 第 2.6.2节—TS、 POD、 FAR、 和 Bias—针对 四个 降水阈值。 这里， "周围 全部数据 参数调优" 表示 我们的 原始 基于 LightGBM 的后处理 模型， 而 "加权 Tweedie" 表示 加权 Tweedie 模型。

表7 成对t检验和威尔科克森符号秩检验结果，比较LightGBM"around all tune"模型（主模型）与温度及风速的基线模型。表格报告了主模型和基线模型的平均绝对误差、绝对误差配对差异的标准差（$\mathrm{SD}_{\mathrm{diff}}$）以及相应的单侧$p$值。所有检验均在$N = 682$个（305个配对样本）上进行。

变量	Baseline	$\mathrm{MAE}_{\mathrm{main}}$	$\mathrm{MAE}_{\mathrm{base}}$	标准差diff	p_t	p_W
T温度	MSM	0.9888	1.3964	1.0840	0.000	0.000
Temp温度	MSMG	0.9888	1.0275	0.7675	0.000	0.000
风速	MSM	0.7549	1.4186	1.3530	0.000	0.000
风速	MSMG	0.7549	0.9035	0.6980	0.000	0.000

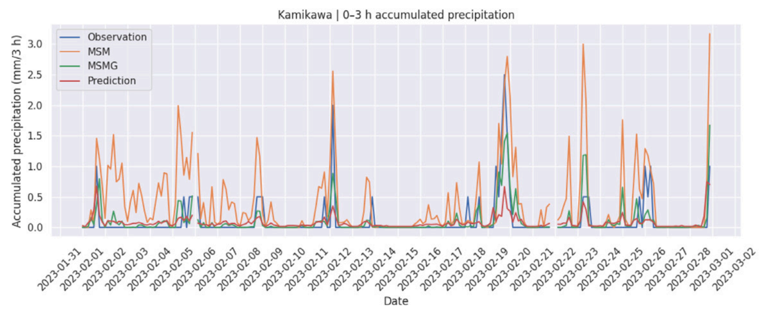
表格 8
降水的基于事件验证指标，汇总所有地点和预报时效。较高的TS和POD表示更好的检测技能，较低的FAR表示较少的虚警，偏差值越接近1表示频率匹配越好。其中，"around all tune"表示原始的基于LightGBM的后处理模型。

阈值(mm)	Model	TS	POD	FAR	Bias
1.0	MSM	0.4311	0.6373	0.4287	1.1156
1.0	MSMG	**0.4555**	0.6558	**0.4014**	**1.0956**
1.0	周围网格数据 全部数据 参数调优	0.4293	0.6931	0.4699	1.3075
1.0	加权Tweedie	0.4435	**0.7005**	0.4527	1.2800
5.0	MSM	0.3226	0.5061	0.5293	1.0753
5.0	MSMG	**0.3344**	0.5014	0.4991	**1.0009**
5.0	周围网格数据 全部数据 参数调优	0.3059	0.3943	**0.4228**	0.6831
5.0	加权 Tweedie	0.3316	**0.5208**	0.5228	1.0912
10.0	MSM	0.2450	0.4084	0.6201	1.0749
10.0	MSMG	0.2485	**0.4122**	0.6152	1.0713
10.0	周围网格数据 全部数据 参数调优	0.1966	0.2356	**0.4571**	0.4340
10.0	加权 Twee die	**0.2520**	0.3979	0.5927	**0.9771**
15.0	MSM	0.1902	0.3189	0.6796	**0.9956**
15.0	MSMG	**0.2096**	**0.3551**	0.6616	1.0492
15.0	周围网格数据 全部数据 参数调优	0.1251	0.1420	**0.4877**	0.2772
15.0	加权 Tweedie	0.1883	0.2719	0.6201	0.7158

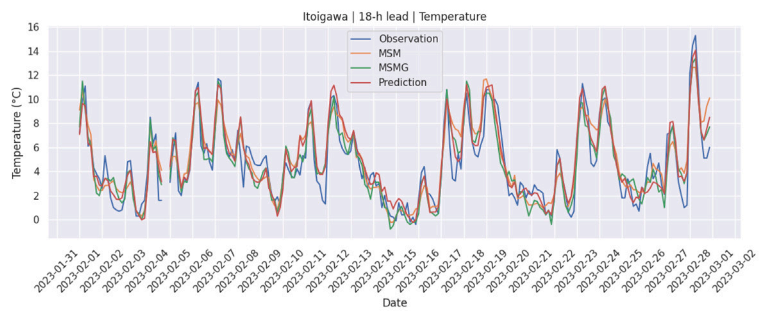
总体而言，MSMG 在对比模型中表现良好。'around all tune'模型在1.0毫米阈值下达到了最高的POD，表明对弱降水事件具有高灵敏度。然而，其FAR和偏差也是最大的，表明它倾向于过度预测降水发生。在5.0毫米及以上阈值，其TS和POD低于MSM和MSMG，偏差在10.0毫米和15.0毫米处远低于1，表明对较强降水事件的检测不足。相比之下，加权Tweedie模型在10.0毫米处改善了TS和偏差，并在5.0毫米处提高了POD。尽管其整体性能仍略低于MSMG，但在不同降水阈值间表现出了更好的平衡。

3.2. 特征选择 方法 的 比较

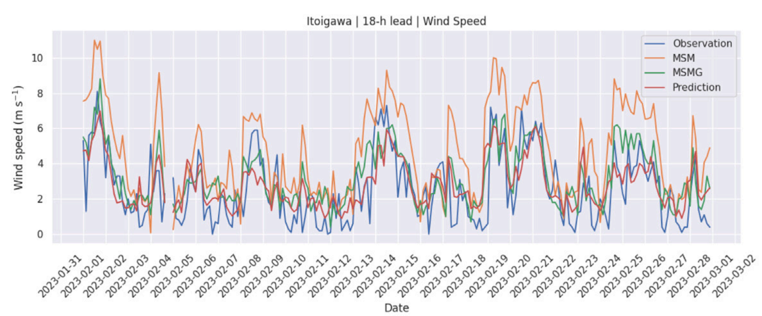
在此，我们于关原的一个地点进行了实验（类似于参数调优），以评估第2.3节提出的三种特征选择方法（FS1 - FS3）中选定特征数量的影响。对于仅使用最接近预测点的网格数据（单网格模型）以及包含周围网格数据（全部数据）的两种情况，分别应用了三种特征选择方法，并通过逐步增加选定特征数量（每次增加10个）来构建模型。图5总结了针对每种特征数量的验证数据的均方根误差k。结果表明，随着特征数量增加，精度趋于提高，但当特征数量超过一定阈值后趋于稳定。这表明相关分析后进一步进行特征选择并不能提升精度。



(a) Accumulated precipitation for the 0–3 h forecast interval at Kamikawa



(b) 18-h lead temperature at Itoigawa



(c) 18-h lead wind speed at Itoigawa

Fig. 4. Examples of prediction results for the test data in February, compared with MSM, MSMG, and observed values. Panel (a) shows accumulated precipitation for the 0–3 h forecast interval, while panels (b) and (c) show temperature and wind speed at the 18-h forecast lead time, respectively.

3.3. Results by location and prediction time

3.3.1. Location-based results

Fig. 6 summarizes the RMSE on test data for each observation location. Our post-processing model consistently outperformed MSM and showed comparable or slightly better performance than MSMG across most sites. For precipitation, relatively large RMSE values were observed in the southwestern part of Japan, including Ashimine and Uchinoura. This is likely associated with the frequent occurrence of typhoons and strong convective rainfall events in these regions. In contrast, the lowest precipitation RMSE values were found in inland and mountainous areas such as Kusatsu and Nobeyama.

For wind speed, extremely high RMSE values in MSM were observed in island and coastal areas such as Niijima and Oma. Our model significantly reduced these errors, achieving RMSE values comparable to those in other regions. Overall, the regional pattern of RMSE reduction indicates that the post-processing model effectively mitigates the biases and noise that are spatially correlated with topography and surrounding meteorological conditions.

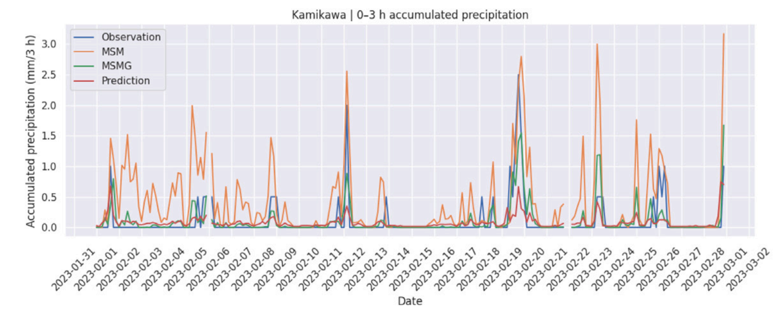
As shown in Table 9, the selected *around_all_tune* model achieved the lowest RMSE across all terrain types and variables. The improvement was most pronounced in mountainous and island areas, where spatial heterogeneity in meteorological conditions is large. This demonstrates that incorporating spatial context from surrounding grids improves model robustness under complex terrain and coastal influences.

Table 9
Average RMSE by terrain category and model. The three terrain categories are defined as follows: **mountain** (Kusatsu, Kamikawa, Nobeyama, Uwanoko-gen, Minamiaso), **island** (Ashimine, Niijima), and **coastal/plain** (Ishikari, Oma, Imabari, Kamaishi, Tsujido, Itoigawa, Uchinoura, Sekigahara, Saitama, Kuwana, Sakai).

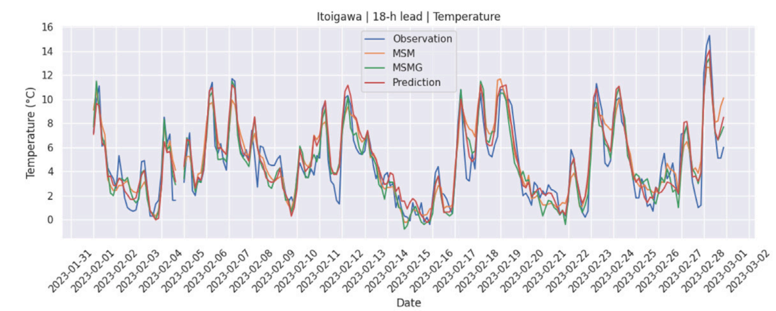
Region	around_all_tune			MSM			MSMG		
	Precip.	Temp.	Wind	Precip.	Temp.	Wind	Precip.	Temp.	Wind
coastal/plain	1.982	1.343	1.002	2.454	1.793	1.864	2.400	1.363	1.211
island	2.951	1.003	1.576	3.492	1.654	2.550	3.349	1.106	1.647
mountain	2.369	1.398	0.822	2.630	1.804	1.337	2.605	1.508	1.038

3.3.2. Prediction-time-based results

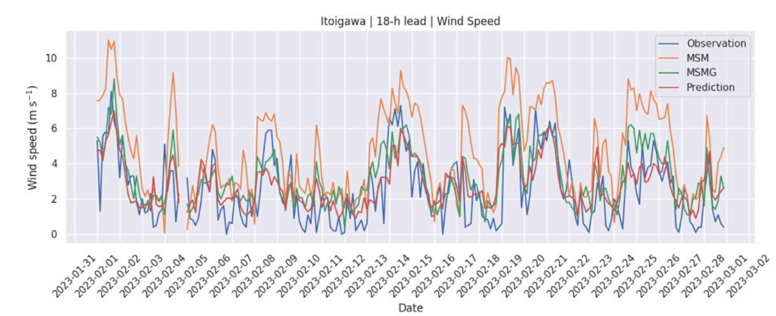
Fig. 7 summarizes the RMSE on test data for each prediction time. In terms of RMSE, the selected around_all_tune model outperformed both MSM and MSMG at almost all forecast lead times. For all methods, RMSE increased as prediction time became longer, which is a common



(a) Accumulated precipitation for the 0–3 h forecast interval at Kamikawa



(b) 18-h lead temperature at Itoigawa



(c) 18-h lead wind speed at Itoigawa

图4. 二月测试数据的预测结果示例，与MSM、MSMG及观测值对比。图(a)显示0-3小时预报时段的累积降水量，图(b)和(c)分别显示18小时预报时效的温度和风速。

3.3. 按 地点 和 预测 时间 的结果

3.3.1. 基于地点 的结果

图6总结了每个观测站点的测试数据的均方根误差。我们的后处理模型始终优于MSM，并且在大多数地点与MSMG表现相当或略优。对于降水，日本西南部（包括Ashimine和Uchinoura）的均方根误差值相对较大。这可能与这些地区频繁发生台风和强对流降水事件有关。相比之下，在Kusatsu和Nobeyama等内陆和山区，降水均方根误差值最低。

对于风速，MSM在岛屿和沿海地区（如Niijima和Oma）出现了极高的均方根误差。我们的模型显著降低了这些误差，使均方根误差达到与其他地区相当的水平。总体而言，均方根误差降低的区域模式表明，后处理模型有效减小了与地形及周围气象条件空间相关的偏差和噪声。

如表9所示，周围网格数据_全部数据_参数调优模型在所有地形类型和变量中取得了最低的均方根误差。改善在山地和岛屿地区最为显著，这些地区的空间

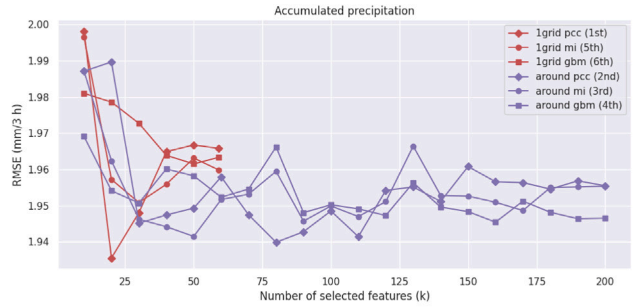
气象条件的异质性 较 大。 这 表明 ， 融入 来自 周围 网格 的 空间 上下文 信息 可 提升 模型 在 复杂 地形 和 海岸 影响 下的鲁棒性。

表9 按地形类别和模型划分的平均均方根误差。三个地形类别定义如下：山地（Kusatsu, Kamikawa, Nobeyama, Uwanoko- gen, Minamiaso）、岛屿（Ashimine, Niijima）和沿海/平原（Ishikari, Oma, Imabari, Kamaishi, Tsujido, Itoigawa, Uchinoura, Sekigahara, Saitama, Kuwana, Sakai）。

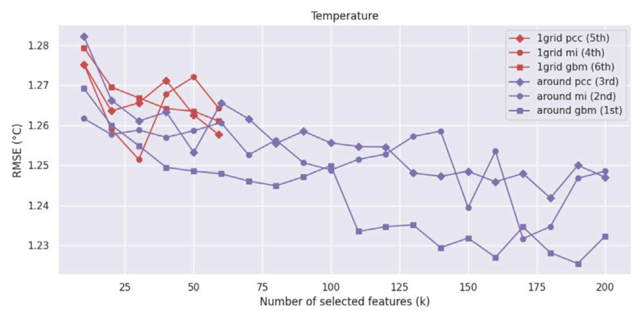
区域	周围_全部数据_参数调优	MSM	MSMG
	降水 气温 风速 降水 气温 风速 降水 气温 风速		
沿海/平原 1.982	1.343 1.002 2.454	1.793 1.864 2.400	1.363 1.211
岛屿 2.951	1.003 1.576 3.492	1.654 2.550 3.349	1.106 1.647
山地 2. 369	1.398 0.822 2. 630	1.804 1.337 2. 605	1.508 1.038

3.3.2. 基于预测时间的结果

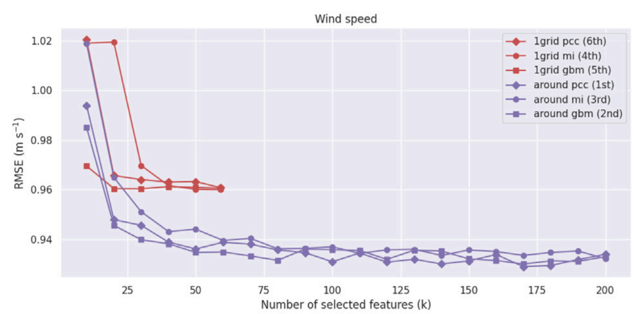
图7总结了每个预测时间在测试数据上的均方根误差。在均方根误差方面，选定的周围_全部 参数调优模型在几乎所有预报时效中都优于MSM和MSMG。对于所有方法，均方根误差随着预测时间变长而增加，这是常见的



(a) Precipitation



(b) Temperature



(c) Wind Speed

Fig. 5. Changes in accuracy based on the number of selected features for the feature selection methods on validation data in Sekigahara. Results are averaged over prediction times.

characteristic of numerical weather prediction (NWP) models, where forecast accuracy declines with lead time. A similar trend was observed in our post-processing results, indicating that the residual errors of the base NWP propagate into the corrected predictions as forecast uncertainty accumulates. Nevertheless, the selected around_all_tune model maintained lower RMSE than MSM and, in many cases, MSMG across forecast horizons, confirming its stability over time.

3.3.3. Scatter plots against observations

Fig. 8 compares observations with the model outputs at Sekigahara during the test period. For readability, all lead times are pooled and a random subset of 150 samples is plotted.

3.4. Site-dependent behavior of the weighted tweedie model

The effect of the weighted Tweedie model was strongly site dependent. To illustrate this point, we compare two representative examples:

Table 10
Site-wise precipitation verification metrics for two representative cases. Higher TS and POD indicate better detection skill, while Bias values closer to 1 indicate better frequency matching.

Site	Model	Threshold (mm)	TS	POD	Bias
Kamikawa	MSMG	1.0	0.3802	**0.5501**	**0.9933**
	around all tune	1.0	0.3817	0.5311	0.9367
	weighted Tweedie	1.0	**0.3863**	0.5319	0.9276
	MSMG	5.0	**0.2447**	**0.3897**	**0.9464**
	around all tune	5.0	0.1541	0.1852	0.3492
	weighted Tweedie	5.0	0.1264	0.1488	0.3057
	MSMG	10.0	**0.1278**	**0.2802**	**0.9973**
	around all tune	10.0	0.0267	0.0275	0.0687
	weighted Tweedie	10.0	0.0471	0.0522	0.0962
	MSMG	15.0	**0.0667**	**0.1703**	**0.9725**
	around all tune	15.0	0.0000	0.0000	0.0000
	weighted Tweedie	15.0	0.0055	0.0055	0.0055
Saitama	MSMG	1.0	**0.4391**	0.6383	**1.0919**
	around all tune	1.0	0.4150	0.6911	1.3563
	weighted Tweedie	1.0	0.4330	**0.7160**	1.3696
	MSMG	5.0	**0.3549**	**0.4994**	**0.9065**
	around all tune	5.0	0.2308	0.2521	0.3444
	weighted Tweedie	5.0	0.3429	0.4959	0.9420
	MSMG	10.0	0.1935	0.3231	0.9923
	around all tune	10.0	0.1493	0.1538	0.1846
	weighted Tweedie	10.0	**0.3395**	**0.4962**	**0.9577**
	MSMG	15.0	0.2048	**0.3675**	1.1624
	around all tune	15.0	0.0678	0.0684	0.0769
	weighted Tweedie	15.0	**0.2600**	0.3333	**0.6154**

Kamikawa, where the improvement was limited, and Saitama at the 18-h lead time, where the weighted Tweedie model showed clearer improvement in the time-series comparison.

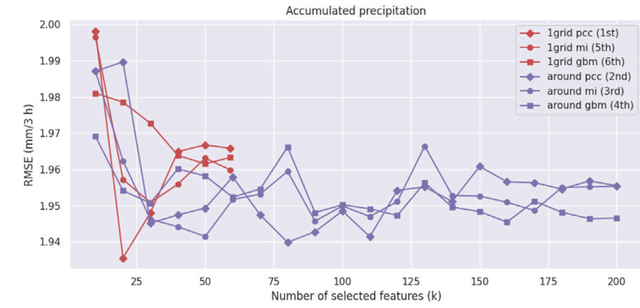
Fig. 9(a) shows the 0–3 h accumulated precipitation at Kamikawa. In this case, the weighted Tweedie model remained close to the original "around all tune" model, and the difference between the two predictions was small for many events. This behavior is consistent with the site-wise aggregated verification results summarized in Table 10. At Kamikawa, the weighted Tweedie model produced only a slight change at the 1.0 mm threshold and did not improve TS at 5.0 mm, although TS and POD were slightly improved at 10.0 and 15.0 mm. Overall, however, the weighted Tweedie model still remained clearly below MSMG for this site, indicating that the modification was not sufficient to overcome the difficulty of precipitation forecasting at this site.

By contrast, Fig. 9(b) shows a case at Saitama for the 15–18 h accumulated precipitation, where the weighted Tweedie model more clearly modified the predicted peaks and, for several events, tracked the observed precipitation more closely than "around all tune". This behavior is also reflected in the site-wise aggregated results in Table 10. At Saitama, the weighted Tweedie model improved TS relative to "around all tune" at all four thresholds and substantially increased POD at 5.0, 10.0, and 15.0 mm, while also moving Bias much closer to 1 than the original model at 5.0 and 10.0 mm. At 15.0 mm, the weighted Tweedie model still showed the best TS among the compared models, although its Bias remained below 1. These results suggest that the weighted Tweedie model can improve event-oriented behavior for some sites, especially when the original model underdetects moderate to heavy precipitation.

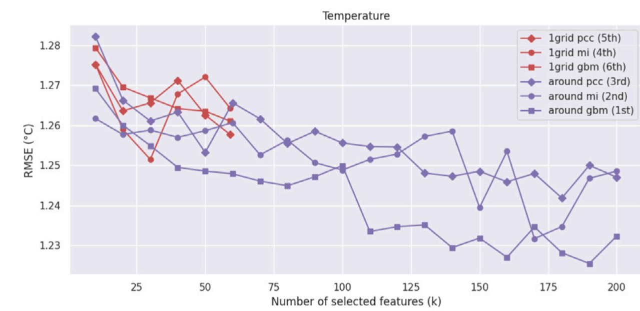
Taken together, these examples indicate that the weighted Tweedie model is useful mainly as a site-dependent adjustment rather than as a uniformly superior replacement for the original LightGBM model. This observation is consistent with the need for site-wise hyperparameter tuning discussed in Section 2.7.

4. Conclusion

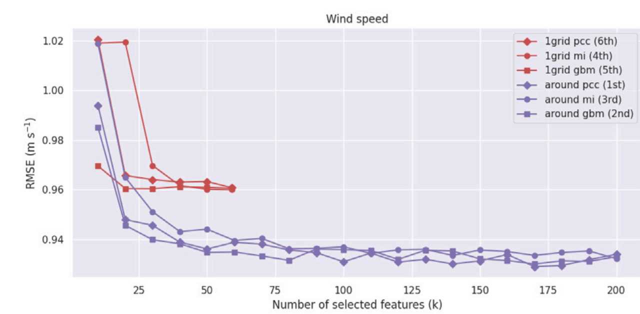
In this study, we developed machine learning-based post-processing models for precipitation, temperature, and wind speed using observation data and MSM data across 18 locations in Japan, including plains,



(a) Precipitation



(b) Temperature



(c) Wind Speed

图 5. 基于 所选特征数量 的 准确率 变化 ， 针对 特征选择方法 在 验证数据 上 于 Sekigahara 的 结果。 结果 在 预测时间 上 取平均。

数值天气预报(NWP)模型的特征，即预报精度随提前时间下降。在后处理结果中也观察到了类似的趋势，表明随着预报不确定性累积，基础NWP的残差会传播到修正后的预测中。尽管如此，选定的周围_全部 参数调优模型在所有预报时效中均保持比MSM更低的均方根误差，并且在许多情况下低于MSMG，证实了其随时间推移的稳定性。

3.3.3. 与观测值的散点图

图 8 比较了测试期间Sekigahara的观测值 与 模型输出。 为了可读性， 所有预测时间 被合并 ，并随机选取了150个样本进行绘图。

3.4. 加权*Tweedie*模型的站点依赖行为

加权 Tweedie 模型 的 效果 具有 强烈 的 站点 依赖性 。为了说明 这 一点 ， 我们 比较 两个 具有 代表性的 例子:

表 10 两个代表性案例的站点降水验证指标。 较高的TS 和POD 表示更好的检测能力， 而偏差值越接近1表示频率匹配越好。

Site	Model	阈值 (mm)	TS	POD	Bias
Kamikawa	MSMG	1.0	0.3802	**0.5501**	**0.9933**
	周围网格数据 全部数据 参数 调优	1.0	0.3817	0.5311	0.9367
	加权 Tweedie	1.0	**0.3863**	0.5319	0.9276
	MSMG	5.0	**0.2447**	**0.3897**	**0.9464**
	周围网格数据 全部数据 参数 调优	5.0	0.1541	0.1852	0.3492
	加权 Tweedie	5.0	0.1264	0.1488	0.3057
	MSMG	10.0	**0.1278**	**0.2802**	**0.9973**
	周围网格数据 全部数据 参数 调优	10.0	0.0267	0.0275	0.0687
	加权 Tweedie	10.0	0.0471	0.0522	0.0962
	MSMG	15.0	**0.0667**	**0.1703**	**0.9725**
	周围网格数据 全部数据 参数 调优	15.0	0.0000	0.0000	0.0000
	加权 Tweedie	15.0	0.0055	0.0055	0.0055
Saitama	MSMG	1.0	**0.4391**	0.6383	**1.0919**
	周围网格数据 全部数据 参数 调优	1.0	0.4150	0.6911	1.3563
	加权 Tweedie	1.0	0.4330	**0.7160**	1.3696
	MSMG	5.0	**0.3549**	**0.4994**	**0.9065**
	周围网格数据 全部数据 参数 调优	5.0	0.2308	0.2521	0.3444
	加权 Tweedie	5.0	0.3429	0.4959	0.9420
	MSMG	10.0	0.1935	0.3231	0.9923
	周围网格数据 全部数据 参数 调优	10.0	0.1493	0.1538	0.1846
	加权 Tweedie	10.0	**0.3395**	**0.4962**	**0.9577**
	MSMG	15.0	0.2048	**0.3675**	1.1624
	周围网格数据 全部数据 参数 调优	15.0	0.0678	0.0684	0.0769
	加权 Tw Tweedie	15.0	**0.2600**	0.3333	**0.6154**

Kamikawa （改进 有限 ） 和 Saitama 在 18小时 的 预报时效 （ 该 处 加权 Tweedie 模型 在 时间序列 比较中 显示出 更明显 的 改进 ） 。

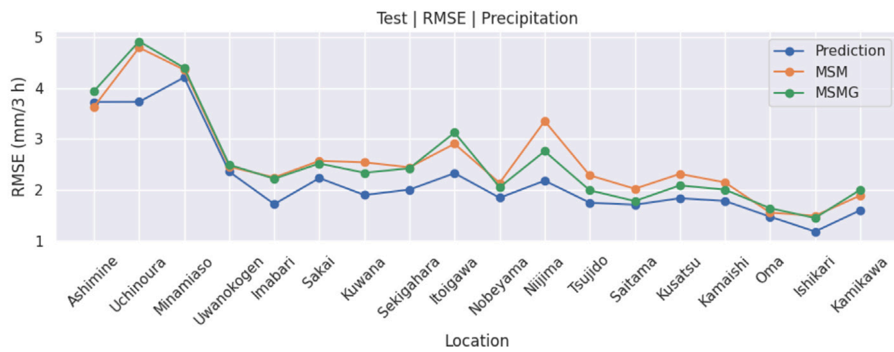
图9(a)显示了Kamikawa地区0–3小时累积降水量。在这种情况下，加权Tweedie模型保持接近原始的"around all tune"模型，两个预测之间的差异在许多事件中很小。这种行为与表10总结的站点聚合验证结果一致。在Kamikawa，加权Tweedie模型仅在1.0毫米阈值处产生了微小变化，在5.0毫米处未改善临界成功指数，尽管在10.0和15.0毫米处临界成功指数和POD略有改善。但总体而言，加权Tweedie模型在该站点仍明显低于MSMG，表明修改不足以克服该站点降水预报的困难。

相比之下，图9(b)显示了埼玉县15–18小时累积降水量案例，其中加权Tweedie模型更明显地修改了预测峰值，并且在多个事件中比"around all tune"更紧密地跟踪观测到的降水。这种行为也反映在表10的站点聚合结果中。在埼玉县，加权Tweedie模型在所有四个阈值上改善了相对于"around all tune"的临界成功指数，并在5.0、10.0和15.0毫米处显著提高了POD，同时在5.0和10.0毫米处将偏差更接近于1。在15.0毫米处，加权Tweedie模型在比较模型中仍显示出最佳临界成功指数，尽管其偏差仍低于1。这些结果表明，加权Tweedie模型可以改善某些站点的事件导向行为，特别是当原始模型低估中等至强降水时。

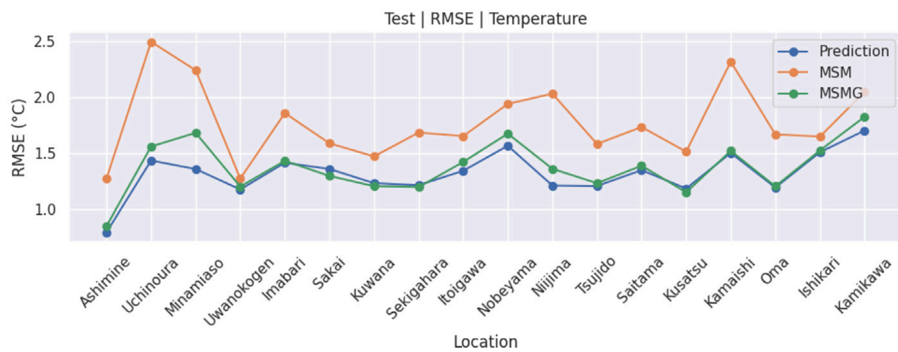
综合来看，这些例子表明加权Tweedie模型主要用作站点依赖调整，而不是作为原始LightGBM模型的统一优越替代方案。这一观察结果与第2.7节讨论的站点级超参数调整需求一致。

4. 结论

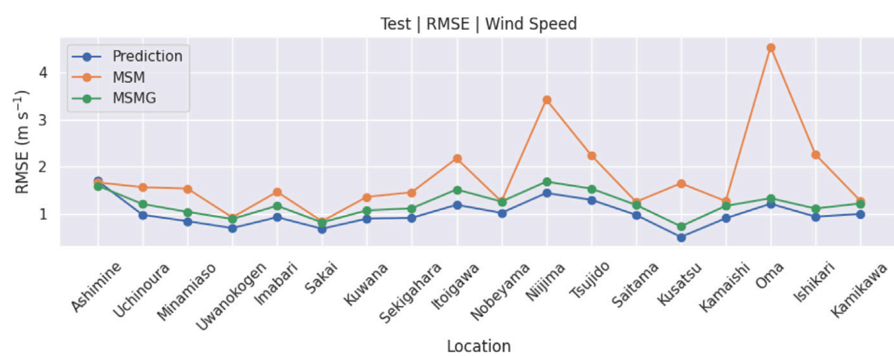
在本 研究 中， 我们 开发了 基于机器学习的 后处理 模型 ， 用于 降水、 温度、 以及 风速 预测 ， 使用了 观测数据 和 MSM 数据 ， 覆盖了 日本 的 18个 地点， 包括 平原，



(a) Precipitation



(b) Temperature

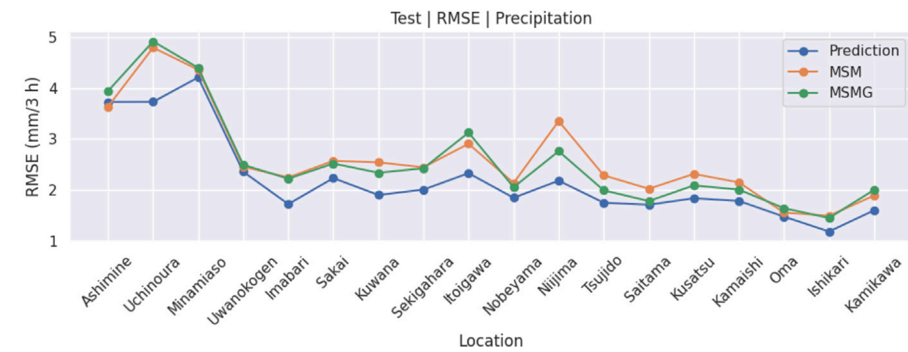


(c) Wind Speed

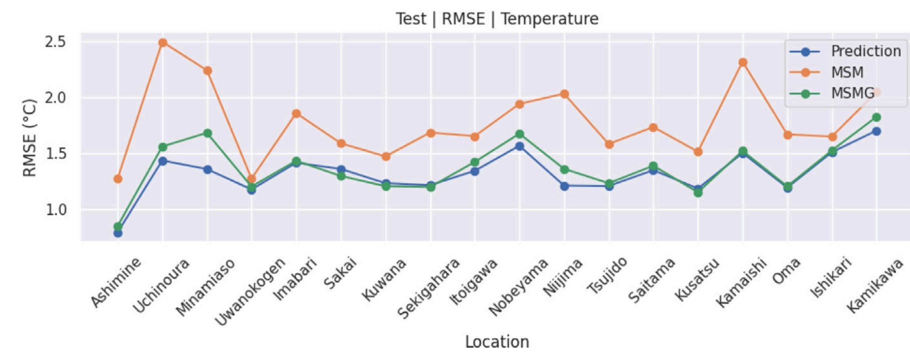
Fig. 6. RMSE by location on test data. Results are averaged over prediction times.

mountainous areas, and islands. The main findings are summarized as follows:

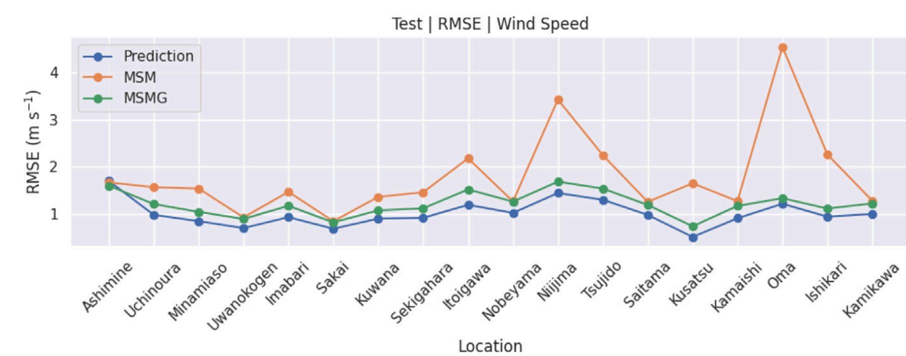
1. The use of surrounding grid data together with correlation-based feature selection improved prediction accuracy compared with using only the single nearest grid. This result supports the effectiveness of incorporating spatial information from surrounding grids while controlling feature dimensionality through feature selection. In contrast, further feature selection beyond this correlation-based reduction did not provide additional improvement in our experiments.
2. In our experimental setting, the LightGBM-based models achieved lower RMSE than the tested neural-network baselines, including the reproduced CNN baseline, and also generally achieved lower RMSE than the JMA guidance product MSMG. However, because the neural-network comparison was limited to specific baseline implementations, the present results do not exclude the possibility that stronger deep-learning architectures could perform better.
3. For precipitation, the results showed that optimizing only standard regression-oriented objectives is not necessarily sufficient, especially when the forecasting goal is to improve the prediction of moderate and heavy rainfall events. To address this issue, we introduced Tweedie-based loss functions and event-weighted training strategies. These approaches improved event-oriented behavior, including better detection of stronger rainfall events



(a) Precipitation



(b) Temperature



(c) Wind Speed

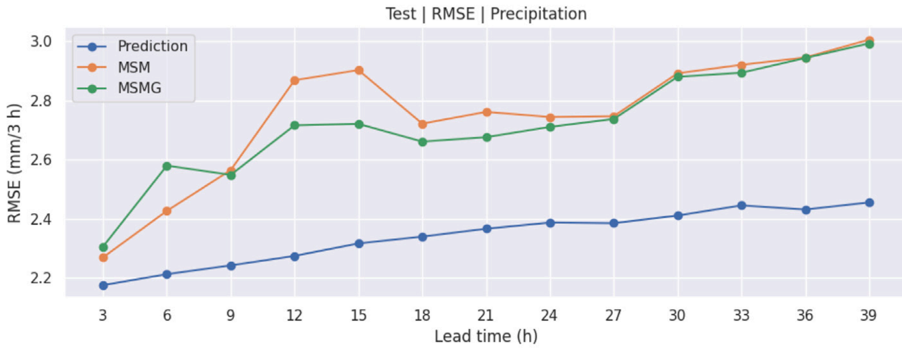
图 6. 均方根误差 按 位置 在 测试 数据上。 结果 是 平均 在 预测 时刻上。

山地 地区，以及 岛屿。 主要 研究结果 总结 如下 ：

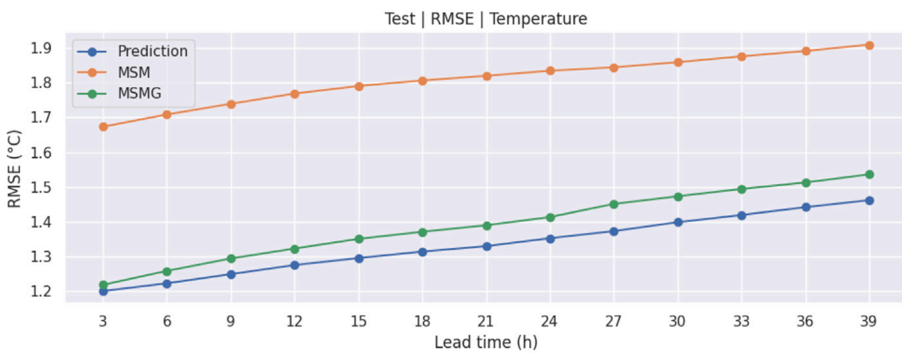
1. 使用周围网格数据结合基于相关性的特征选择，相比仅使用单个最近网格，提高了预测精度。这一结果支持了在通过特征选择控制特征维度的同时，融入周围网格空间信息的有效性。相比之下，在我们的实验中，在此基于相关性的降维基础上进一步进行特征选择并未带来额外改进。

2. 在我们的 实验 设置 中， 基于LightGBM的 模型取得了 比 测试 的 神经网络 基线更低的 RMSE， 包括复现的CNN基线在内，并且通常也比JMA指导产品MSMG获得了更低的RMSE。然而，由于神经网络比较仅限于特定的基线实现，当前结果并不排除更强的深度学习架构可能表现更好。

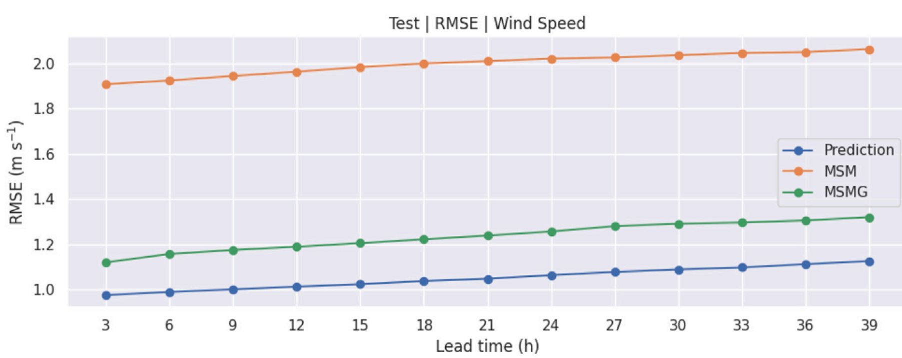
3. 对于降水，结果显示仅优化标准回归导向的目标并不一定足够，特别是当预测目标是改善中等和强降雨事件的预测时。为解决此问题，我们引入了基于Tweedie的损失函数和事件加权训练策略。这些方法改善了事件导向行为，包括更好地检测较强降雨事件。



(a) Precipitation

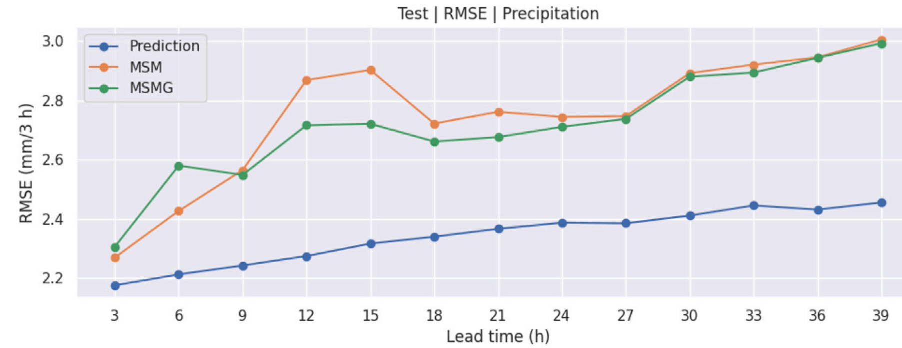


(b) Temperature

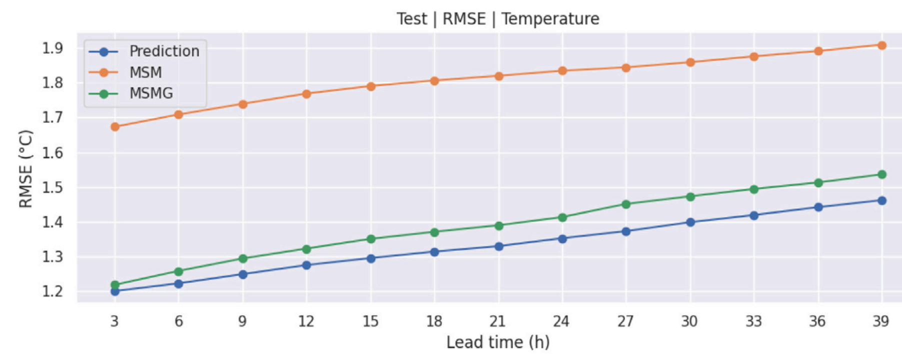


(c) Wind Speed

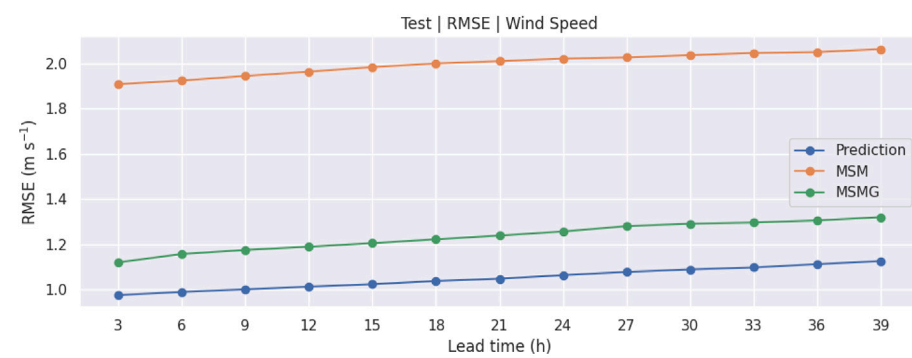
Fig. 7. RMSE by prediction time on test data. Results are averaged over locations.



(a) Precipitation

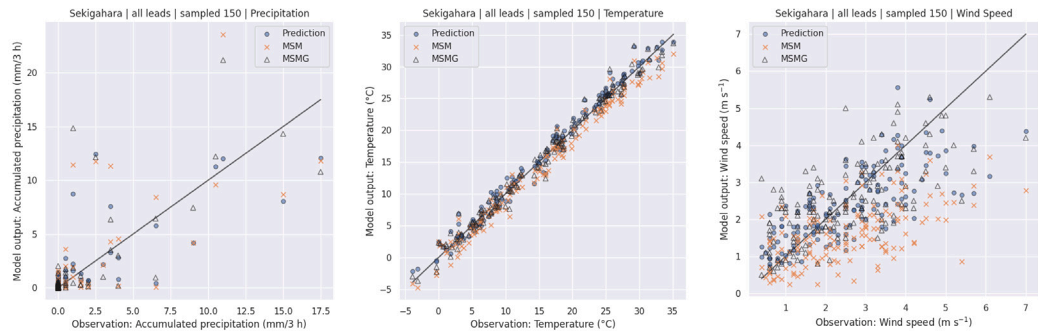


(b) Temperature



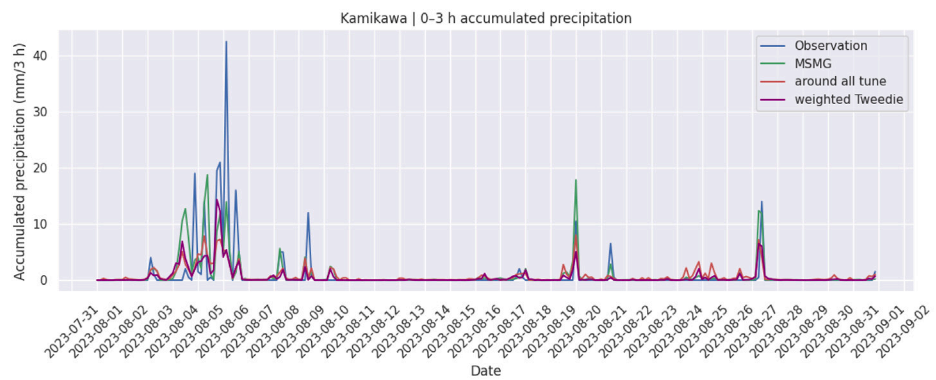
(c) Wind Speed

图 7. 均方根误差 按 预测 时间 在 测试 数据上。 结果 被平均 到 各地点。

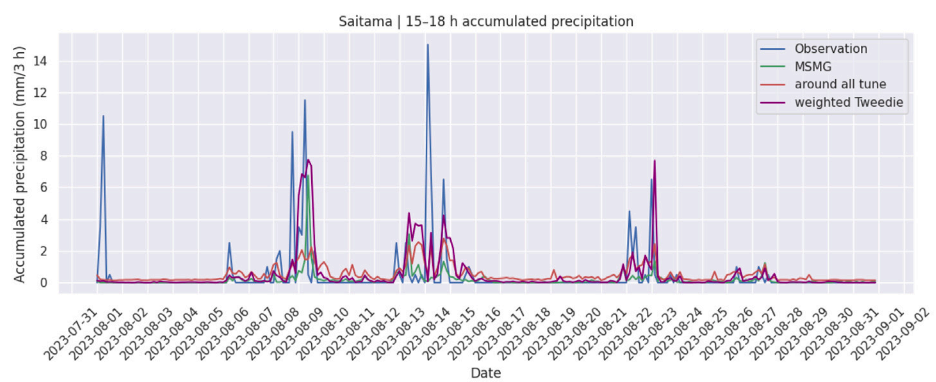


(a) Precipitation (b) Temperature (c) Wind speed

Fig. 8. Scatter plots at Sekigahara on the test period (all lead times pooled). Each panel shows observations versus model outputs for the same randomly sampled points.



(a) Accumulated precipitation for the 0–3 h forecast interval at Kamikawa



(b) Accumulated precipitation for the 15–18 h forecast interval at Saitama

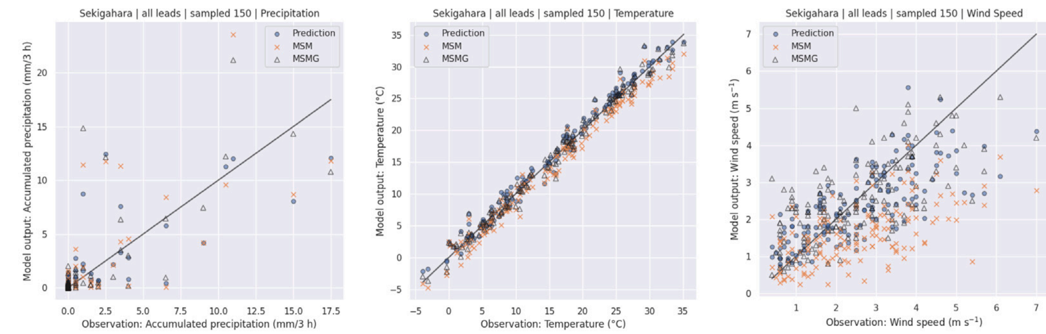
Fig. 9. Comparison of precipitation forecasts for two representative cases. Panel (a) shows Kamikawa, where the weighted Tweedie model remained close to the original "around all tune" model and the improvement was limited. Panel (b) shows Saitama, where the weighted Tweedie model showed improved event-oriented behavior for several precipitation events.

at some sites and thresholds. However, the gains were not uniform across locations and conditions, indicating that further improvement is still needed. This site dependence suggests that inter-station differences related to topography and site characteristics, such as latitude, longitude, and altitude, may need to be considered more explicitly in future work.

To complement the representative examples and pooled results shown in the main text, detailed verification results by site and forecast lead time are provided in the Supplementary Material. These include RMSE and ME of MSM, MSMG, and "around all tune" for precipitation, temperature, and wind speed, as well as precipitation event-based metrics (TS, POD, FAR, and Bias), including those of the weighted Tweedie model.

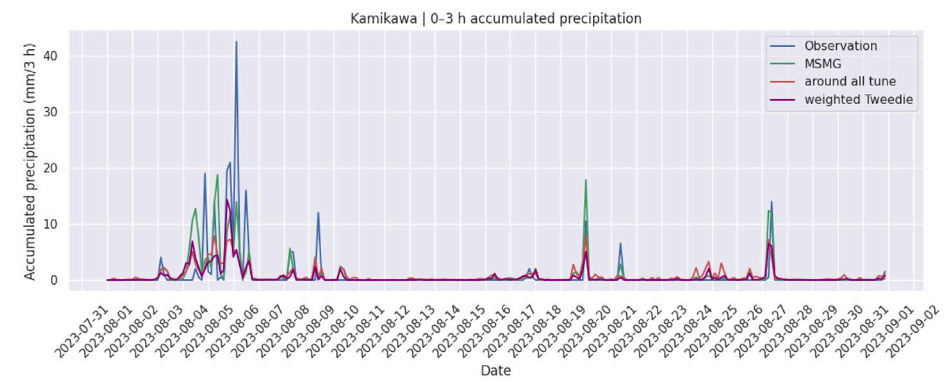
CRediT authorship contribution statement

Kazuma Iwase: Writing – original draft, Methodology, Data curation, Conceptualization. **Tomoyuki Takenawa:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Conceptualization.

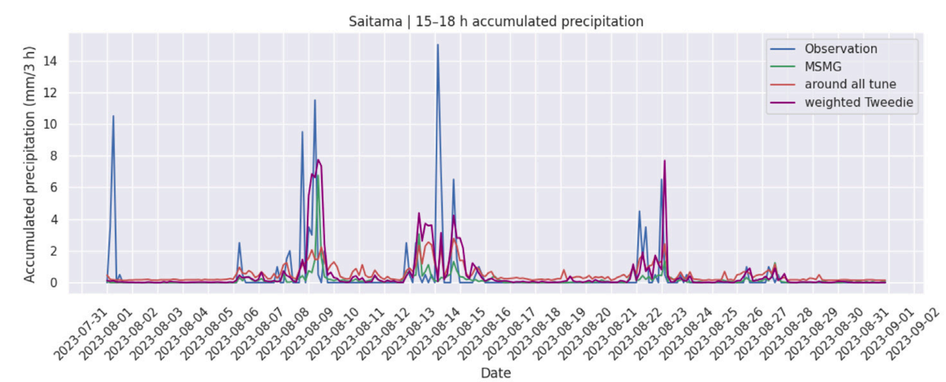


(a) Precipitation (b) Temperature (c) Wind speed

图. 8. 在Sekigahara测试期间（所有提前时间合并）的散点图。 每个面板显示同一随机采样点的观测值与模型输出对比。



(a) Accumulated precipitation for the 0–3 h forecast interval at Kamikawa



(b) Accumulated precipitation for the 15–18 h forecast interval at Saitama

图9. 两个代表性案例的降水预报对比。图(a)为Kamikawa，加权Tweedie模型保持接近原始“around all tune”模型，改进有限。图(b)为Saitama，加权Tweedie模型在多次降水事件中显示出改进的事件导向行为。

在某些地点和阈值下。然而，改进并非在所有地点和条件下一致，表明仍需进一步改进。这种地点依赖性表明，与地形和站点特征（如纬度、经度和海拔）相关的站点间差异可能需要在未来工作中更明确地考虑。

为了补充主文中展示的代表性示例和汇总结果，附录中提供了按站点和预报提前时间划分的详细验证结果，包括降水预报的均方根误差（RMSE）和平均误差（ME），涉及MSM、MSMG和'around all tune'模型。

温度 、 风速 以及 基于 降水的 事件型 指标 （ 临界成功指数（TS） 、 探测概率（POD） 、 虚报率（FAR） 和 偏差（Bias） ）， 包括 加权 Tweedie 模型的结果。

CRediT 作者署名 贡献 声明

Kazuma Iwase: 写作 – 原始草稿，方法论，数据策管，概念化. **Tomoyuki Takenawa:** 写作 – 审阅与编辑，写作 – 原始草稿，监督，资源，项目管理，方法论，概念化.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this manuscript, the authors used Chat-GPT to assist with code editing and language proofreading. All AI-assisted outputs were reviewed and revised by the authors as needed, and the authors take full responsibility for the content of the manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.atmosres.2026.109037.

Data availability

The code used in this study is available at: https://github.com/ttakenawa/PostMSM. Due to data licensing restrictions, the repository provides a synthetic/derived dataset for demonstration, which may not exactly reproduce the paper results.

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声明 生成式 AI 与 AI辅助 技术 在 稿件 准备 流程 中的 使用

在 本 手稿 的 准备 过程中， 作者 使用 了 Chat-GPT 来 辅助 代码 编辑 和 语言 校对 。 所有 AI辅助 输出 均已 由 作者 根据需要 进行 了 审查 和 修改 ， 作者 对 手稿 的 内容 承担 全部数据 责任 。

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附录 A. 补充 数据

补充 材料 与 本文 相关 的 补充 材料 可 在线 获取 于 https://doi.org/10.1016/j.atmosres.2026.109037。

数据 可用性

本研究使用的代码可在以下位置获取：https://github.com/ttakenawa/PostMSM。由于数据许可限制，该仓库提供了一个合成/衍生数据集用于演示，可能无法精确重现论文结果。

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